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Simulated Laser Triangulation with Focus on Subsurface Scattering

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Simulerad lasertriangulering med fokus på subsurface scattering

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Abstract

Laser triangulation is a contact-free and optical measurement technique that can be used to derive optical surface properties such as reflectance and scattering, in addition to the shape of the object. A laser triangulation system consists of a laser, a camera and an object to be measured. These system parts can be simulated to create a set-up that is more flexible, time efficient and requires less work force than a physical laser triangulation system. The thesis work was performed at SICK IVP AB, using Blender, and it included two measurement objects: a wooden spruce plank and a blister package. Methods for realistic simulation of each system part were explored. Approaches for including subsurface scattering in the simulated measurement objects were also examined. Lastly, the thesis answers how a simulated laser triangulation system can be compared with a practical system in order to evaluate the realism of the simulation.

Practical laser triangulation sessions were performed for each measurement object to obtain ground truth data. Three methods for laser line simulations were implemented: reshaping the built-in light sources of Blender, creating a texture projector and approximating a Gaussian beam as a light emitting volume. The camera simulation was based on the default camera of Blender together with settings from the physical camera. Three approaches for creating wood material were tested: procedural texturing, using microscopic image textures to create 3D-material and UV-mapping high resolution photograph onto the geometry. The blister package was simulated with one material for the pills and another for the semi-transparent plastic packaging. A stand-alone Python script was implemented to simulate anisotropic/directed subsurface scattering of a point laser in wood. This algorithm included an approach for creating vector fields that represented subsurface scattering directions. Three post-processing scripts were produced to simulate sensor noise, blurring/blooming of the laser line and lastly to apply simulated speckle patterns to the laser lines. Sensor images were simulated by rendering a laser line projected onto a measurement object. The sensor images were post-processed with the three mentioned scripts. Thousands of sensor images were simulated, with a small displacement of the measurement object between each image. After post-processing, these images were combined to a single scattering image. SICK provided the algorithms needed for laser centre extraction as well as for scattering image creation.

All laser methods worked and gave sufficiently realistic results, except for the estimation of a Gaussian beam that was not finalised. This method could perhaps be completed using a different renderer. The speckle pattern and sensor noise simulations resulted in images that visually and statistically resembled ground truth data. Parameter tweaking for each script and system part could potentially increase the realism of the simulation. As long as the measurement object does not scatter anisotropically, it is possible to simulate a full laser triangulation system using Blender.

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1 Introduction

Laser triangulation is a measuring technique that can be used to produce many types of measurements with high accuracy, from thickness, width and height to outer and inner diameters, profiles and distances [1]. These measurement are derived from sensor images and by combining different measurements, three dimensional data can be obtained. In addition to 3D data, laser triangulation can also be used to extract reflectance and scattering information from the measurement objects. System parts that are needed in order to perform laser triangulation are the object to be measured, a laser emitter, a camera with a laser sensor and movement.

SICK IVP AB, further referred to as SICK, manufactures instruments for laser triangulation, among other things. They are offering their measuring techniques for various uses, such as for quality assurance, identification and inspections. Several industries have need for precise production and therefore also for highly accurate measurements. The construction industry is one example and companies developing medical instruments are another.

By analysing the pixel intensity of the sensor images produced by laser triangulation, different optical properties of the measured object can be obtained. One such property is the amount of *subsurface scattering*, which describes how a material, depending on its transparency, absorbs, reflects and transmits light. Subsurface scattering can be used for identification and classification cases, by distinguishing materials by their different optical properties.

Laser triangulation is essential for the wood industry; precise measurements, quality control and creating prototypes are large parts of the industry. Subsurface scattering in wood is especially interesting due to the varying properties of the material. Light, for instance, travels differently in and outside of branches or knots. The amount of knots, rot and resin - important factors for quality control of wood - can all be measured using laser triangulation with subsurface scattering.

Another use case for laser triangulation is to make sure that blister packages are ready for shipping. Detecting defects and presence/absence of pills is relatively easy in transparent blister packages. Once these packages are coloured and instead semi-transparent, more complex methods are needed for quality assurance. Laser triangulation can be used for this purpose; the amount of subsurface scattering in the triangulation result can be used to tell whether or not each blister contains a pill. This is due to the fact that light behaves differently depending on if it reaches a pill or the bottom of the blister package. Laser triangulation in-

cluding subsurface scattering can be used in the pharmaceutical industry to make sure that coloured blister packages are defect free and contain the right amount of pills.

1.1. Motivation

Determining set-ups for laser triangulation is a time consuming process that requires work force as well as a proper physical object to be measured. Testing new ideas, rethinking or changing the plan is often inevitable and something that can lead to a great loss in time and labour. Factors that make laser triangulation hard to set up and time demanding include finding suitable measurement objects that contain all interesting variations. Further time consuming tasks consist of moving large measurement objects around and having to re-calibrate the relations between the system parts after each movement.

Simulating laser triangulation has many benefits, saving time and work force. The algorithms for the practical/physical triangulation can be developed and improved without expensive and time consuming practical tests. Prototypes for new products can easily be created, altered and fine-tuned in a simulation before they are physically produced. Training data for various algorithms can be generated through simulated laser triangulation as well. Simulations enable for flexible and easy accessible work; with a computer as a tool new ideas can be tested, developed and scrapped with less loss in time, material and work compared to practical laser triangulation.

One way of simulating is through a rendering software. Blender is free, open source and allows for plug-ins and add-ons, making it suitable for this thesis [2].

In order to achieve realism in simulated laser triangulation, physical artefacts or phenomena must be included. Examples of such artefacts are optical phenomena (interference, refraction), aberrations in the camera/optics (astigmatism, distortions) as well as variations in the laser (speckle, varying thickness/focus/wavelength). Since subsurface scattering can be used in many cases, by several industries, this phenomenon is the focus of this thesis. SICK's latest 3D camera sensor supports a technique for easy measuring of subsurface scattering. This allows for comparisons of simulated and physical subsurface scattering.

1.2. Aim

The aim of the work is to examine methods for simulating laser triangulation with high realism. These methods entail modelling of each separate part of a laser triangulation system: camera (with sensor), laser, measurement object (geometries with materials) and movement. Further explorations are to be made on approaches for including subsurface scattering in the measurement objects where the phenomenon naturally occurs. In order to evaluate the realism of the simulation, the simulated results are to be compared with results from practical laser triangulation.

1.3. Research questions

The work aims to answer the following research questions:

1. How can realistic line lasers be simulated in the rendering software Blender?
2. How can realistic wood and blister package materials that include subsurface scattering be simulated in Blender?
3. How can the realism of simulated laser triangulation be evaluated through comparison with practical laser triangulation?

1.4. Delimitations

Simulation of laser triangulation will be implemented for the two measurement objects of raw spruce wood and coloured blister packages. Methods for post-processing of the sensor images will not be explored extensively. These methods include how to extract the laser line, how to measure subsurface scattering and how to combine the simulated sensor images to scattering images. Instead SICK's library of implemented algorithms will be used for the post-processing, saving time for simulating the different system parts. Calibration of the laser triangulation set-up, including camera calibration, is also considered out of the scope of this thesis.

2 Theory

Laser triangulation is a contact-free and optical measurement method. The term laser triangulation originates from the triangle that is created between the laser, the camera and the object to be measured, see *Figure 2.1*. For scanning 3D-objects laser triangulation is the most used contact-free measurement method, partly because of the fact that it cannot damage the measurement object, but also since the method is time efficient [3], accurate and allows for relatively easy system construction [4].

When a laser is projected onto a measurement object it is deformed and reflected off of the object's surface. The reflected laser light will then contain information about the surface from which it was reflected. The fixated camera captures this reflected light, the object or laser is moved and the method is repeated. Some situations require the entire measurement object to be scanned, whereas in other cases only the interesting parts of the object are triangulated. In order to produce full 3D-models, for example, the whole object has to be covered by the laser. For measuring certain surface properties however, it might be sufficient to scan only selected areas. The result of a laser triangulation session is a fully/partly scanned object, with the wanted surface information captured by the camera [5][6].

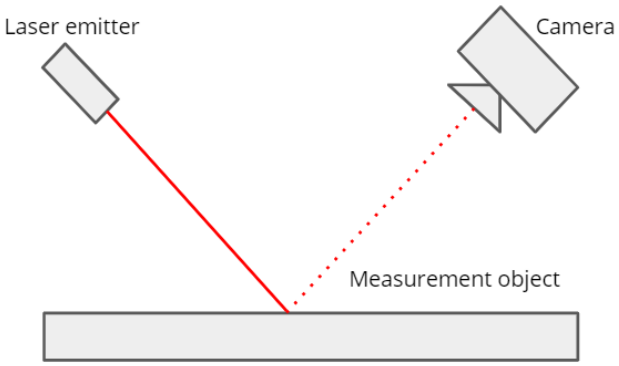


Figure 2.1: Illustration of a basic laser triangulation system including a camera, a laser and a measurement object.

To obtain 3D-models and/or data, an algorithm is needed to define what pixels in the sensor images that are the reflected laser light. Once the laser line has been extracted, the remaining bright pixels of the image can be classified as lit by subsurface scattering. Using another algorithm for measuring the amount of subsurface scattering in a sensor image, the scattering of each point of a measurement object can be calculated. These algorithms are, as mentioned in the delimitation, outside the scope of this thesis. An overview of the steps of a laser triangulation session is seen in the following list:

1. Set up laser triangulation scene with measurement object, laser and camera.
2. Project laser onto the measurement object.
3. Capture a sensor image with the camera, depicting the laser on top of the measurement object.
4. Extract the laser line from the sensor image from step 3.
5. Measure surface properties in the sensor image from step 3, using the extracted laser line from step 4.
6. Move the laser or the measurement object and repeat step 2-5 until the desired area of the measurement object has been scanned by the laser.
7. Combine data from several sensor images to obtain 3D-data.

Theory of each part of a laser triangulation system is presented below, followed by a section on methods for improving the measurement accuracy of laser triangulation. Lastly, approaches for simulating each system part are introduced.

2.1. System parts

The needed instruments for laser triangulation are, as previously mentioned, a laser, a camera, an object to be measured and also the movement of the laser or the geometry. All parts except for the movement are illustrated in *Figure 2.1* above.

The following sections will cover how movement can be included in a laser triangulation system. Further, basic laser theory is presented together with an introduction to the laser property *speckle*. The main parts of digital cameras are described with sections on camera sensors and frame rate. Interesting properties of the objects to be measured are discussed as well as the phenomenon of subsurface scattering.

2.1.1. Movement

In order to measure or scan an object the laser has to traverse the whole object surface. This can be done either by sequentially moving the laser or the object [7]. The movement of the laser and/or geometry does not have to be performed along an axis. Rotation can also be used as movement, for instance similarly to how Babar et al. rotate the measurement objects in a full circle to obtain a 3D-model [5]. Another common scanning method is to use *fast steering mirrors* (FSMs) to move the laser point or line over the object surface. The laser is then aimed at a mirror that can be sequentially rotated. Depending on the tilt/angle of this mirror, the laser is projected onto different positions on the measurement object [8].

When choosing a movement method factors to consider are the shape of the object, the wanted result format and the type of laser. For example, a point laser requires more movement than a line laser to cover the entire measurement object. When planning the movement the collision of system parts and camera/laser occlusion should also be considered [3]. There are naturally physical boundaries of a system set-up; the system parts cannot be placed at the

same position and the camera or laser cannot see or hit objects that are occluded or blocked from their view.

2.1.2. Laser

Laser is, unlike most common light sources, a *coherent* light source. This entails that the light is represented by a single wavelength with a constant phase and frequency. An *incoherent* light source, in comparison, has varying phases and could also consist of several wavelengths. *"Normal"* white light is an example of an incoherent light, consisting of all wavelengths in the visual range. Coherent light sources, like lasers, are often used when scanning objects. This is mainly because coherent light can be subject to effects caused by *diffraction* [9].

There are different types of laser emitters or projection devices that can be used in laser triangulation, where the most common ones are point and line lasers. Babar et al. also include dual line laser as part of the conventional laser triangulation set-ups [5]. Point lasers can be transformed to shape line lasers. This is done by emitting a point laser through a lens, either a cylindrical lens or a so called Powell lens, fanning out the point into a plane [9]. The different laser shapes (line or point) have separate advantages making them suitable for various use cases.

With line lasers the direction or rotation of the line relative to the measurement object has to be taken into consideration, whereas point lasers are uniform and their relative rotation have no impact on the measurement results. According to Zhang et al. the accuracy of the triangulation is higher when using a point laser, especially when scanning materials with varying properties, also called heterogeneous materials [10]. This is partly because a laser point covers less surface area than a laser line in one projection, allowing for more detailed information to be captured. Furthermore, the point laser is less likely to cover several characteristics of a material at once. Since the point laser will most often interact with a single surface property in each projection, the laser power and camera settings can be adjusted for every different surface characteristic. Point lasers can also be used to measure directional scattering, which is not possible with a line laser [11]. Scanning a larger object would be significantly more time consuming with a point laser than with a line laser, which is why most laser triangulation systems use a line laser.

Laser properties

Depending on the colour and material of the measurement object, different wavelengths may work better than others. This is due to the fact that different colours reflect and absorb different wavelengths. The various types of laser triangulation results can also benefit from different coloured lasers. If a triangulation system uses a red laser, surfaces with high red colour components will yield more accurate scanning results than any other colours [12] [13].

Another important characteristic of lasers, similar to their wavelength, is their intensity or power. Laser power also has to be adjusted to fit the colour and material of the measurement object. Dark surfaces demand stronger lasers in order to be seen; this is because they absorb the majority of the incident light. The opposite is true for brighter surfaces: they need lower laser intensities to be seen since they already reflect most of the incoming light [12].

Depending on the type of laser emitter, different types of laser properties will be present. Most laser beams can be approximated to have a Gaussian intensity distribution along the transverse plane (cross section). Gaussian beams cannot be focused on a single point, which causes them to never be perfectly sharp. Two further important properties of the Gaussian beam is the beam waist together with the beam width. The beam waist describes the point where the laser is most concentrated (most in focus). The width of the laser beam will vary depending on the distance to the beam waist [9].

Speckle patterns

Whenever using lasers as light sources on *optically rough* surfaces, diffraction can cause the occurrence of speckle, which is an important factor for laser measurements [14]. Optically rough means that the surface has spatial differences that causes the reflected light rays to travel separate distances, making them phase shifted. These delayed or phase shifted rays can either reinforce (*constructive interference*) or cancel each other out (*destructive interference*), creating a pattern of brighter and darker areas in the reflected laser light [9]. These speckle patterns can either be *objective* (viewed directly in free-space) or *subjective* (imaged through another diffracting element, like a digital camera) [14][9]. An example of a speckle pattern is seen in *Figure 2.2* below.

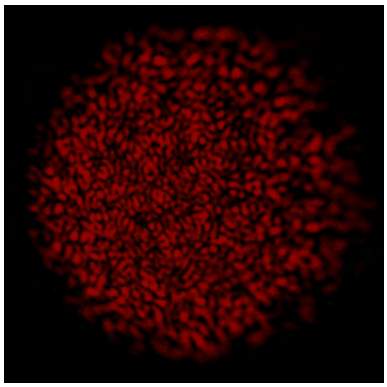


Figure 2.2: Photograph of a speckle pattern, captured by Christophe Finot, accessed from Wikimedia Commons 2021-06-16, showing a noisy pattern created by a red coherent light source.

2.1.3. Camera

The basic laser triangulation set-ups use a digital camera with optics suitable for the measurement scene [3]. Some camera and optics characteristics that affect the accuracy of laser triangulation are the distance between lens centre and laser (or target) in the different directions, the lens' *focal length* as well as the tilt of the lens [15].

Lenses and stops are used to control the essential light passage throughout a camera. These lenses and stops are often spherically shaped, made out of plastic or glass and placed on a shared axis. One of the most important stops is the *aperture*, which regulates the size of the hole through which light passes to the sensor or image plane. *Shutter speed* controls for how long this aperture should be kept open and together with the aperture, they constitute the base of a camera's exposure system [16]. Newer camera systems have refrained from mechanical shutters. Instead of a physical stop opening and closing, the light sensitivity of the sensor pixels is turned on and off. Rolling and Global shutters are the main types of electronic shutters. Global shutters are commonly used in various imaging applications, to avoid known artefacts and distortions of the rolling shutters [17].

Moving the lenses of a camera does not only affect the light passage, but it also alters the *field of view (FoV)* and at which distance the camera is focused [16]. The field of view, *field angle*, or *angle of view*, describes the entrance angle of the incoming light [18]. Assuming a lens focused at infinity, the field of view is defined in accordance with equation 2.1 [19].

$$FoV = 2 \times arctan\left(\frac{\text{sensor size}(d)}{2 \times \text{focal length}(f)}\right) \tag{2.1}$$

For lenses that assume infinite focus, the focal length/focal distance is the distance between the lens and the sensor, see *Figure 2.3*. From equation 2.1, it is seen that the focal length

and the FoV are inversely proportional. Setting a longer focal length shifts the lens further away from the sensor, causing the FoV to decrease, see *Figure 2.3*. A larger focal length, along with the smaller FoV, results in a magnification of the image [3][16]. Assuming infinite focus or not, the focus of a camera always depends on the distance between the sensor and the lens; if the target of a photograph is moved closer to the camera, the distance between sensor and lens must be increased in order for the target to remain in focus [18].

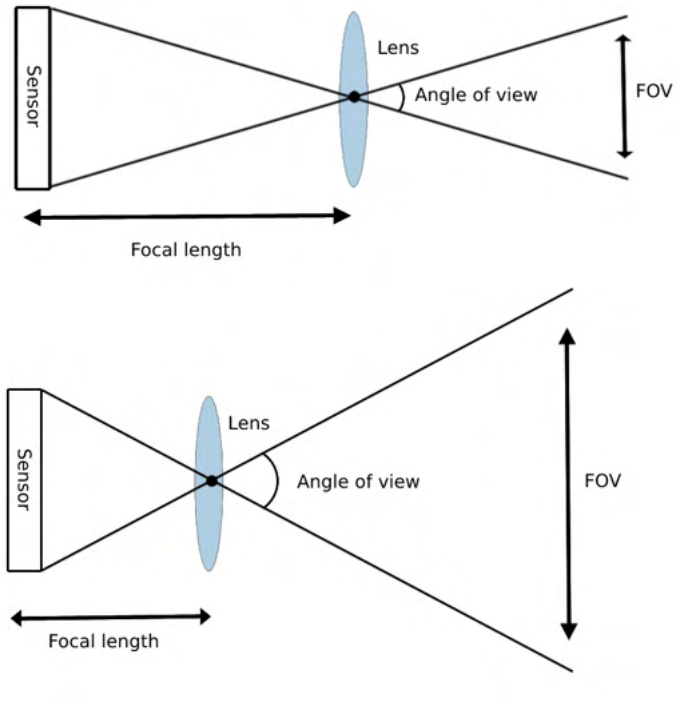


Figure 2.3: Illustration of the relation between the camera parameters field of view (FOV) and focal length assuming a lens focused at infinity; shorter focal lengths yields wider FOV's and vice versa.

The *f-number (f#)* describes the diameter of the aperture as a function of focal length. More in detail, the f-number represents the amount of light that travels through the objective to the sensor per unit of time as the aperture is opened. The f-number is given by the ratio f/n, see equation 2.2, where f is the previously mentioned focal length/distance and n is the luminosity of the objective or optics [18].

$$f\# = \frac{f}{n} \implies n = \frac{f}{f\#} \tag{2.2}$$

Looking at equation 2.2, it can be seen that a longer focal distance will not only cause a smaller entrance angle (FoV), but also a bigger f-number. Large f-numbers equals small aperture diameter, due to the parameters being multiplicative inverses, and vice versa. A small aperture (caused by a large f-number) will cause less light to be gathered in the objective, leading to a darker image [18]. Similarly, larger apertures will let through more light and result in a brighter image.

During laser triangulation, the measurement object and laser line or point should be within the area of focus. To ensure this, the correct focal length, or distance between the lens and the sensor, must be chosen. Since the focal length has an impact on the aperture size (via f-number), which in turn controls the exposure, there is a trade-off between the level of sharpness and the brightness of the image.

Diffraction occurs as light passes through the aperture stop. A large amount of diffraction will result in less sharpness in the final image [18], thus making this effect unwanted. Mohammadikaji et al. state that the amount of diffraction can be decreased but never fully removed with larger aperture openings (smaller f-number) [14]. They further present that larger aperture diameters also reduce the size of speckles in the laser, which could be desirable in some cases. However, larger apertures are not all good; using a large aperture (and a low f-value) also means using a short focal length. A short focal length means that an entire object might not fit within the focal plane, causing parts further away to be out of focus.

Quintana et al. agree with Mohammadikaji et al. that smaller f-values (larger apertures and smaller focal lengths) have advantages. However, they also present that larger f-values, either caused by larger focal lengths or smaller apertures, have benefits as well. For example, they state that optical distortions, mostly introduced by imperfections in the camera, often can be reduced with a larger focal length. This would however, as stated before, also decrease the FoV. Aberrations are harder to avoid and correct with shorter focal distances and to reduce aberrations, Quintana et al. therefore suggest using higher f-values [18].

To conclude, bigger aperture sizes (smaller f-number) seems good to reduce diffraction and longer focal lengths (larger f-values) should decrease distortion, but having both large aperture and focal length is not possible. In most digital cameras, the main parameters that can be altered are the f-number, the shutter speed and the *ISO-value*, which will be covered in the next section on *sensors*.

Camera sensor

The sensor is the part of the camera where the analogue 3D-data of the world is transformed into digital 2D-pixels. The theory behind this transformation is beyond the scope of this thesis. CCD and CMOS are the most common types of sensors, with their separate advantages. CCD sensors usually work better in darker scenes, due to them suffering smaller amounts of noise than CMOS sensors. The biggest positive of CMOS sensors is that they enable using a sub-section of the full sensor. Restricting the capture area like this allows for higher maximum frame rates and lower acquisition times [3].

The main characteristics of a sensor are its resolution (density of digitalised surface points) and its size, along with its amount of electronic noise [3]. The bigger the sensor is, the larger the amounts of gathered light and aberrations become. By combining the resolution and size of the sensor, it is possible to derive its pixel size. This pixel size is the main factor affecting the sensor noise; the larger the pixel size, the smaller the amount of noise. Another important parameter is the sensitivity of the sensor, often measured in the ISO-scale. This ISO-sensitivity also affects the extent of noise, where higher ISO-values cause more noise. Low ISO-values means that the sensor is less sensitive to light, thus needing more light for a proper exposure. The sensitivity of the sensor is increased by amplifying the signal that is gathered by the sensor and this may be needed in situations where the shutter speed cannot be further lowered, but the scene is still too dark [18]. 1

Frame rate and scanning speed

The frame rate of a laser triangulation system describes how often an image, frame or *profile*, should be captured. This rate can be described either by time passed or by displacement of the object/laser, for example as number of profiles per second or millimetre.

The frame rate has to be sufficiently high for capturing the necessary details [5] and to keep the scanning time of the measurement objects reasonably low. It should however not be so high that it forces the system into using a too reduced exposure time, which darkens the image. Higher ISO-values might then be needed to brighten the image but this would increase the sensor's electronic noise. Higher number of captured profiles per millimetre/second corresponds to a lower maximum roof for exposure time, possibly resulting

in lower quality profiles. High scanning speeds also demand a fast system for tracking positions and displacements [3]. This results in another trade-off, this time between the speed of the laser triangulation system and the image quality, which is directly connected to the measurement accuracy.

2.1.4. Measurement object

The material of the measurement objects has great impact on the quality of the laser triangulation result. The colour and the roughness of the material have already been mentioned, in section 2.1.2, to affect the laser in various ways (reflecting/absorbing properties and speckle). Another important characteristic is the material's opacity; laser points or lines will not be reflected off of perfectly transparent materials. Instead the light waves will go via the surface of the object, passing right through, to be reflected on an eventual surface underneath/behind the transparent object. This will result in a flawed 3D representation of the object, since the sensor cannot capture any light that is reflected off of the geometry that is to be recreated. For surfaces that are not completely transparent (like most surfaces), the reflected light commonly consists of a combination of reflections from the top and bottom surfaces. Similar flaws, as those caused by transparency, will occur when a material has excessive amount of specular reflection. The laser light will then only be reflected in a specific direction which may either saturate or miss the sensor completely. Fernandez et al. state that the best laser triangulation results are produced using objects with Lambertian materials [12]. Dizeu et al. confirm that laser triangulation will not work on transparent materials, due to their low levels of reflection. They also establish that laser triangulating semi-transparent objects results in lower accuracy in comparison to triangulating fully opaque materials [20].

Subsurface scattering

Scattering is the light transport phenomenon where the propagation (energy transfer) direction is changed due to a participating medium (material affecting light transport) [21]. Subsurface scattering describes how light scatters (changes directions) beneath a surface that is not fully opaque. One part of an incident light ray is directly reflected off the surface (specular reflection). The other part will penetrate the object surface and hit particles within the surface until all energy is lost, or until the light scatters back out of the surface, see *Figure 2.4* [22].

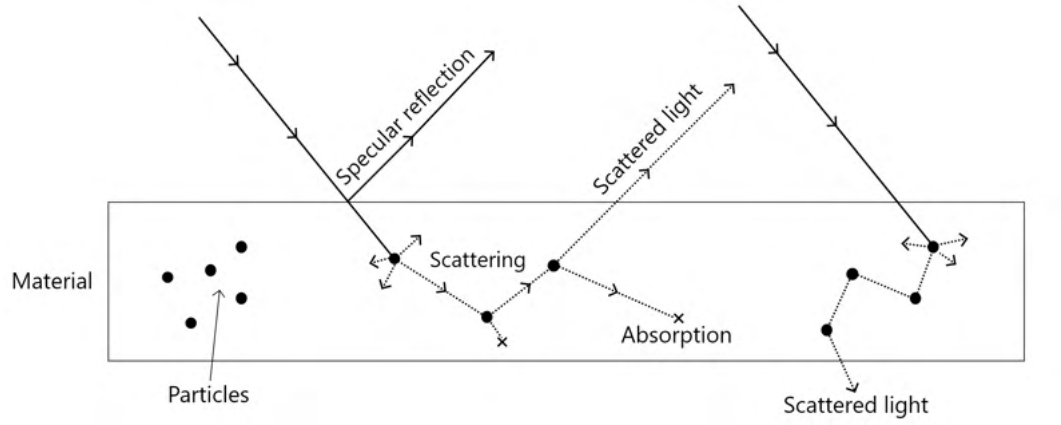


Figure 2.4: Illustration of the subsurface scattering phenomenon occurring in (semi-) transparent materials.

Nicodemus et al. state that the opacity of a surface material determines how far into the material an incoming light ray travels before it is absorbed or scattered back to the surface.

The point at which a scattered light beam exits the surface is rarely the same as the ray's point of incidence. The more translucent the material is, the further apart are the entry and exit points [23]

Subsurface scattering can be *isotropic* or *anisotropic*. Isotropic materials have the same probability of light bouncing in any direction within the material; the direction of the reflected light does not depend on the direction of the incoming light. In anisotropic materials on the other hand, each pair of incoming and outgoing light directions is described by a *phase function*. Different phase functions increase the likelihood of light bouncing in certain directions [24]. A simplified illustration comparing anisotropic and isotropic subsurface scattering is seen in *Figure 2.5*.

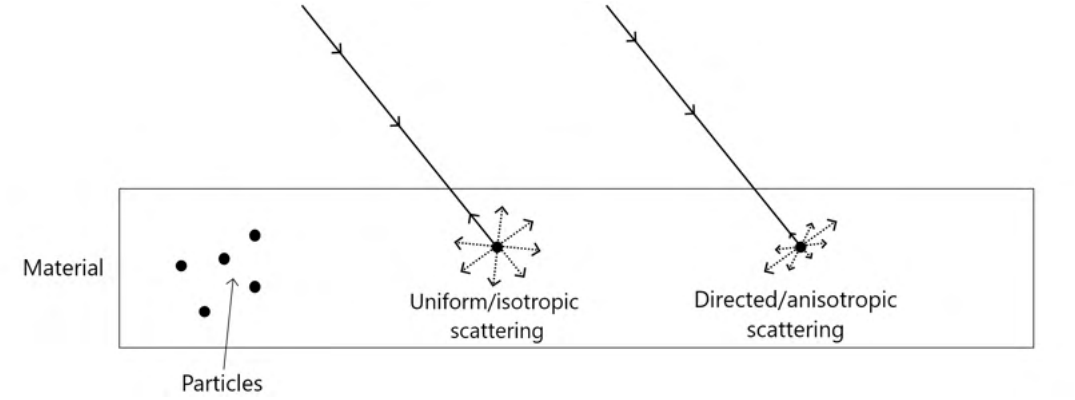


Figure 2.5: Illustration of isotropic subsurface scattering (left incident beam) compared to anisotropic subsurface scattering (right incident beam).

2.2. Improving measurement accuracy

The precision of laser triangulation is mainly affected by the set-up of the optical parts, the laser centroid extraction algorithm and the features of the measurement object surface as well as of the laser source [15]. The image acquisition step can be seen as a bottleneck, since the following image processing steps all depend on the quality of the captured profiles [3].

The detector or camera is placed non-parallel to the laser plane in order to compute the dimensions of the measurement object and thus also obtain its height data [4]. Placing the camera at an angle towards the object and laser plane limits how much of the object that can fit within the camera's plane of focus. In order to maximise how much of the measurement object that is in focus, one can tilt the camera lens according to the *Scheimpflug condition*. The lens is then rotated so that the lens and the image/sensor plane of the camera intersect at a point on the laser plane, see *Figure 2.6*. With the measurement object in focus, the laser line projected onto the object is sharp over the entire field of view [25] [15]. When the lens is rotated, its distance to the sensor is increased, causing the field of view to shrink and the object to be magnified.

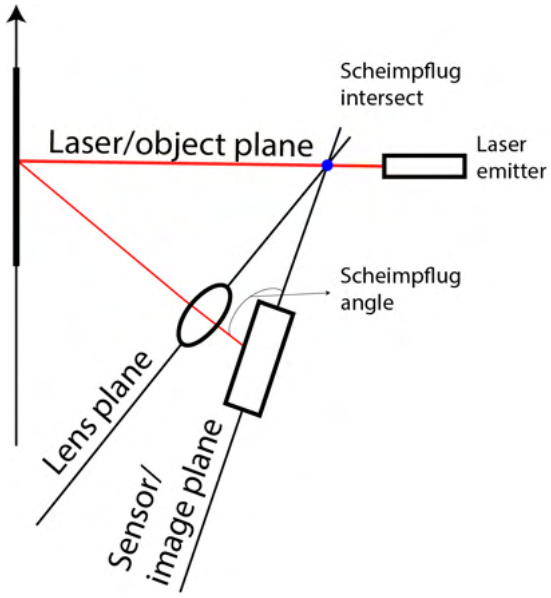


Figure 2.6: Illustration of a laser triangulation set-up that satisfies the Scheimpflug condition.

Different methods can be used to extract the centre of the laser point or laser line. One approach, presented by Babar et al., is to compare all the pixels of the sensor images to a given brightness threshold, flagging the bright pixels as the laser centre [5]. Other methods include using the centre of gravity [14], assuming Gaussian intensity profiles, estimating the intensity spread with linear interpolation or using Taylor series expansion around the peaks [26]. Kienle et al. state that the error caused by the laser centroid extraction algorithm can be reduced by averaging positions from several lasers or by improving the other image processing steps [15]. Further discussion on this topic is out of the scope of this thesis, since it will employ a laser extraction algorithm provided by SICK.

The transparency of the measurement object can cause issues for a laser triangulation system, as discussed in section 2.1.4. If it is not suitable to paint the object, possible workarounds for these cases include altering the laser centre extraction algorithm to focus on the first rather than the strongest intensity peak. This is done since the highest peaks are likely sourced from another surface than the measurement object.

Using lasers with wavelength and power that are appropriate to the specific surface is important for obtaining accurate triangulation results, as mentioned in section 2.1.2. Further laser properties of importance are speckle, focus and occlusion. Points where the laser is out of focus or that are darkened by the destructive interference in the speckle pattern as well as areas occluded from the laser cannot be properly scanned. The choice of laser and using larger apertures to reduce speckle size should be considered when laser triangulating. Lee et al. state that modern laser triangulation set-ups use multiple cameras or lasers to avoid occlusion. However, using more system parts naturally entails a higher production cost [27].

Fernández et al. state in their report that laser triangulation gives the best result when no external lighting is present. If an external lighting is inevitable however, Fernández et al. further conclude through experiments that the Mercury Vapour Lamp (MVL) is the best choice [12]. An alternative method for avoiding ambient or external lighting in laser triangulation is by adopting *selective wavelength filtering*. Optical filters can be applied to the camera so that only light of certain wavelengths are let through to the sensor. Not to be forgotten when using these filters is that since white light contains all wavelengths, some unwanted information might be captured in the presence of white light.

2.3. Simulating laser triangulation

In order to simulate laser triangulation each part of the system has to be modelled accordingly, including movement, laser, camera and the object to be measured.

Mohammadikaji et al. present in their work from 2020, a physically accurate method for simulating optical measurement systems, with a Gaussian beam laser including speckle and with realistic camera sensors [14]. They state that diffraction and *lens aberrations* are some of the most important artefacts to include when simulating laser triangulation.

Another approach for simulating a full laser triangulation system was presented by Beermann et al. in 2018 [6]. They use ray-tracing, a pinhole camera and a laser plane in Hesse's standard/normal form (alternative formulation of a plane, derived from the general plane equation [28]) to virtually measure a cylinder object. The following sections of this chapter exemplify methods for simulating each system part of a laser triangulation set-up, together with important parameters to be considered.

2.3.1. Simulating movement

Since there are several methods for including movement into the laser triangulation systems, there are also several ways of simulating these different kinds of displacements. A simple approach is to sequentially alter the position of the measurement object or the laser until the whole surface has been scanned. Rotating the object a full cycle, in accordance with the previously mentioned method of Babar et al. is also possible (stated in section 2.1.1) [5]. FSMs can be simulated using ray-tracing, as demonstrated by Schlarp et al. [8].

2.3.2. Simulating laser

Different models can be used to approximate and simulate lasers, where one of the most common methods is to model the laser as a *Gaussian beam*. Important properties when simulating realistic lasers as Gaussian beams are shape, intensity, focus and speckle [9].

Simulating a Gaussian point laser requires, according to Bergmann et al., the wavelength of the laser, the size of the beam at focus (beam waist width), the position of the beam waist, the laser direction and the optical power of the laser [9]. To maintain the Gaussian properties when transforming the laser point into a laser line, a cylindrical lens should be used.

A line laser can also be modelled as a plane, for example from the Hessian normal form [6], as mentioned above. Cajal et al. concluded in their work on simulated laser triangulation from 2015 that a higher precision can be achieved if the laser is modelled as a triangle instead of a rectangular plane [3]. This is because a triangle better represents the reality where the laser originates from a single point.

In rendering software, lasers can be simulated by altering the built-in light sources or by creating custom emitters based on physics. Some rendering software have built-in laser light sources as well, such as the LuxCoreRenderer for Blender [29].

Simulating speckle patterns

Speckle patterns can be approximated by simulating noise with statistical properties and behaviour known from speckle theory. Most characteristics of surface materials do not affect speckle statistics; the property of importance is, as described in section 2.1.2, that the surface is optically rough. Duncan and Kirkpatrick introduced a way to create objective and subjective speckle patterns with different intensity distributions (exponential and Rayleigh for example) in their work from 2008 [30]. Their method includes pattern generation for static and moving objects. Goodman et al. reworked Duncan and Kirkpatricks method, integrating theory from other researchers and introduced a technique to generate a stack of patterns to simulate dynamic speckle behaviour [9]. Mohammadikaji et al. state that speckle cannot

be efficiently simulated using current ray-tracing algorithms that are based on geometrical optics [14]. In order to create interference patterns ray-tracers would become complex and require a lot of time, thus making other approaches more suitable.

2.3.3. Simulating camera

A camera can be simulated as a *pin-hole* camera (*camera obscura*), based on perspective projection. Each 3D point within the camera's field of view is projected to a corresponding 2D point on the image/sensor plane [3]. In an ideal pin-hole camera the aperture is assumed to be infinitely small but yet leak enough light to the image plane. Additionally, the focus area is estimated to infinity and blur, lens distortions, light diffraction and other aberrations are not considered. This standard camera is commonly used in simulations, despite not being physically accurate, due to the simplicity of transforming the world 3D points into the pixels on the 2D sensor [6].

In rendering software the built-in camera can be altered to resemble a physical camera through a number of parameters (aperture, focal length and so on), which often also include fields to resemble specific sensors.

Simulating sensor

Mohammadikaji et al. simulated realistic camera sensors, as previously mentioned, and concluded that the most important properties to include are the sensor's dimension (1, 2, 2.5 or 3), size (width and height), pixel size and the amount of noise [14]. The electronic noise of a camera sensor is commonly modelled as Gaussian noise, with given mean and standard deviation. The sensor noise should be included in the simulation prior to the execution of a laser centre extraction algorithm in order to be properly modelled [3].

2.3.4. Simulating measurement objects

A measurement object can be simulated at minimum as geometry with a material. Most rendering software have pre-defined meshes that can be altered to resemble physical objects and then various methods for creating and applying materials. The materials can either describe the surface or the entire volume of the geometry. They can be *homogeneous* and have the same characteristics all over, or *heterogeneous*, meaning that the material has varying properties.

Simulating subsurface scattering

Subsurface scattering determines, as previously explained, how much the light travels within a material before it is absorbed and/or reflected. Jensen et al. stated in their work on subsurface scattering in fur from 2017, that James F. Blinn was the first to introduce subsurface scattering to computer graphics in 1982 [31]. They further stated that many methods have been tested to solve for subsurface scattering, such as path tracing, scattering equations, photon mapping and (anisotropic) dipole solutions.

Blinn explained subsurface scattering in order to synthesise the rings of Saturn [32]. In order for a light ray to be visible after travelling through a material, Blinn stated that there must be no particles in the way along the path of the ray. Modelling this scattering probability with Poisson and inserting it into a function of brightness, Blinn could describe any translucent material. Examples of different types of scattering (Rayleigh and anisotropic for example) was presented together with a few approaches that have been tested to synthesise subsurface scattering, like the *Henyey-Greenstein* method, different solutions with *Lambert's law* as well as weighted averages of several functions.

In 2001, Jensen with colleagues were among the first to use subsurface scattering to enhance realism in computer graphics. They then introduced subsurface scattering as a method

to more realistically simulate diffuse surfaces that were not fully opaque [33]. Subsurface scattering is considered an important phenomena in physics based rendering, to produce proper lighting in (semi-) transparent materials, such as skin, wax, marble and wood. The commonly used lighting model BRDF (bidirectional reflectance distribution function) is often not effective for materials with scattering. Instead, materials with subsurface scattering can be modelled using combinations of BRDF's and BTDF's (bidirectional transmittance distribution function) or combinations of the more complex BSSRDF (bidirectional scattering-surface reflectance distribution function) and BSSTDF (bidirectional scattering-surface transmittance distribution function). The last functions are more complex due to their allowance of the incoming light to leave the object surface at another point than the point of incidence [34].

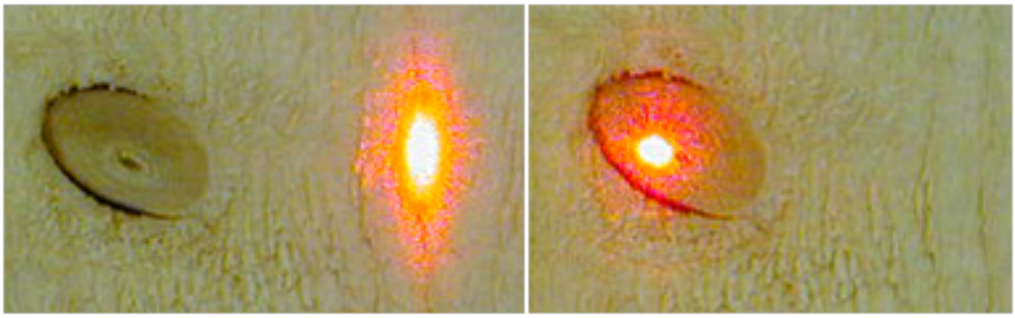
Wrenninge et al. state that most subsurface scattering models assume isotropic behaviour and to simulate anisotropic subsurface scattering, they present a path tracing method [24].

Stam introduces a way to approximate multiple scattering events with a diffusion process [21]. This diffusion method assumes optically thick materials, where scattering is so frequent that the scattered photons are insignificantly dependent on directions. Subsurface scattering can be modelled by performing diffusion on thin material slices representing each subsurface layer. By discretising the scattering directions, Stam further states that anisotropic effects can be simulated.

Simulating raw wood

When light hits a wooden plank it is separated into two components (as in most surfaces): one that is directly reflected off of the surface at the point of incidence and another one that enters the plank, bounces around inside and (possibly) reflects off of the surface at another point. Wood consists of so called *tracheids*, cells that are shaped to follow the fibres of the wood.

Light that hits wood will mostly spread in the direction of the fibres and create an elliptical spread shape, see *Figure 2.7a*. This happens because the tracheids inside of the wood have a high amount of light transmission. This elliptical effect is called the *tracheid effect*. When light hits a branch on a wooden plank the light will have a circular spread instead, see *Figure 2.7b*. This is due to the fact that the fibres inside of a branch or knot are perpendicular to the fibres in the smooth/clear parts of the plank. Light rays incident at knots will therefore mostly travel into the wood, following the fibres. Törmänen and Mäkynen introduced a method to detect branches on a plank through scanning with a point laser and analysing the different spread effects [11].



(a) (b)

Figure 2.7: Point laser on a wooden plank where light scatters elliptically along the tracheids or fibres of the wood (a), compared to the same point laser on the knot of the wooden plank, causing a circular spread shape in (b).Images provided by SICK IVP AB.

About 90% of all the wood cells grow in the same direction. Apart from branches, some trees also have cell clusters that grow perpendicularly to the tracheids. These clusters are called *rays* and they affect the reflectance and scattering at the wood surface. Knots and rays are properties that make wood anisotropic, meaning that it has different features in different directions [35].

Spruces often form clusters of knots, which allow fibre deviations caused by one knot to interfere or overlap with deviations from another. Lukacevic et al. introduce this as an important property that needs to be considered in any fibre modelling algorithm [36].

Wood textures have been modelled and simulated in various ways. Procedural methods, projection of 2D images onto geometries and using BRDF's are a few examples [35]. Lukacevic et al. presented a more high end method for modelling wood in 2019, as they developed 3D-models for fibre directions on and around knots, including an algorithm for knot reconstruction. Three planks of Norway spruce were modelled and evaluated based on whether or not they were suitable for use in quality controls and for assessments of mechanical properties such as stiffness and bending [36].

To picture and analyse scattering data of wood, Törmänen and Mäkynen state that point lasers are preferred. This is mainly due to the fact that a line laser must be placed perpendicular to the tracheids of the wood in order to show fibre direction and scattering [11].

Simulating blister packages

Blister packages are common in the pharmaceutical production industry. They consist of the four components *forming film*, *lidding material*, *heat-seal coating* and *printing ink*, where the first component (forming film) makes up the majority of the package (80-85%). Forming film can be made out of different plastic materials (PVC, polypropylene (PP) and polyester (PET)), aluminium and/or combinations of several materials. Lidding material is composed of different types of aluminium or mixes of aluminium, plastics and paper. 15-20% of blister packages is the lidding material. Heat-seal coating is what combines the forming film with the lidding material and it can be solvent or water based. The last component is the printing ink, which is some type of high quality ink [37].

Forming film is often transparent and without colour, so that the pills can be easily sighted. To protect medicines that are sensitive to light the forming film is sometimes coloured white. One pill that is distributed in coloured blister packages is Ibuprofen Orifarm 400 mg film coated tablet. According to the product resume, this particular blister package consists of PVC (polyvinyl chloride) and aluminium foil [38].

Laser triangulation can be used to discover defects and damages in coloured blister packages. The method can also be used to tell whether or not each blister contains a pill. Defects and damages can be found by looking at the profile of the package, represented in the height maps (2D images with height data in each pixel) that is a result of the measuring technique. By looking at the spread or scattering of the reflected laser light instead, Matthias Johannesson produced images where it was clearly visible which blisters that contained a pill and which did not. This was possible due to the different amounts of absorption and reflection in the semi-transparent blister and the pill [39].

Optical properties of PVC and aluminium foil have not been found to be commonly documented. Yousif et al. presented in 2013 a table of the index of refraction (IOR)/refractive index for pure PVC under light of wavelengths between 300 and 600 nanometres. The IOR was found to increase from just below 1.5 at 300 nm to become static at 1 for all wavelengths longer than or equal to 350 nm [40]. Further information on how light scatters in the main materials of the chosen blister package was not found.

3 Method

The laser triangulation simulation was performed in the rendering software Blender (versions 2.91 and 2.92) [2], by simulating a camera, a laser and geometries with proper materials. An overview of the tested methods for simulation of each system part is illustrated in *Figure 3.1*. Blender's Python API [41] was used together with external Python scripts to implement the system. The Blender scenes were implemented using scripts in order to simplify version control as well as future work. The external Python scripts were used for post-processing of the Blender output.

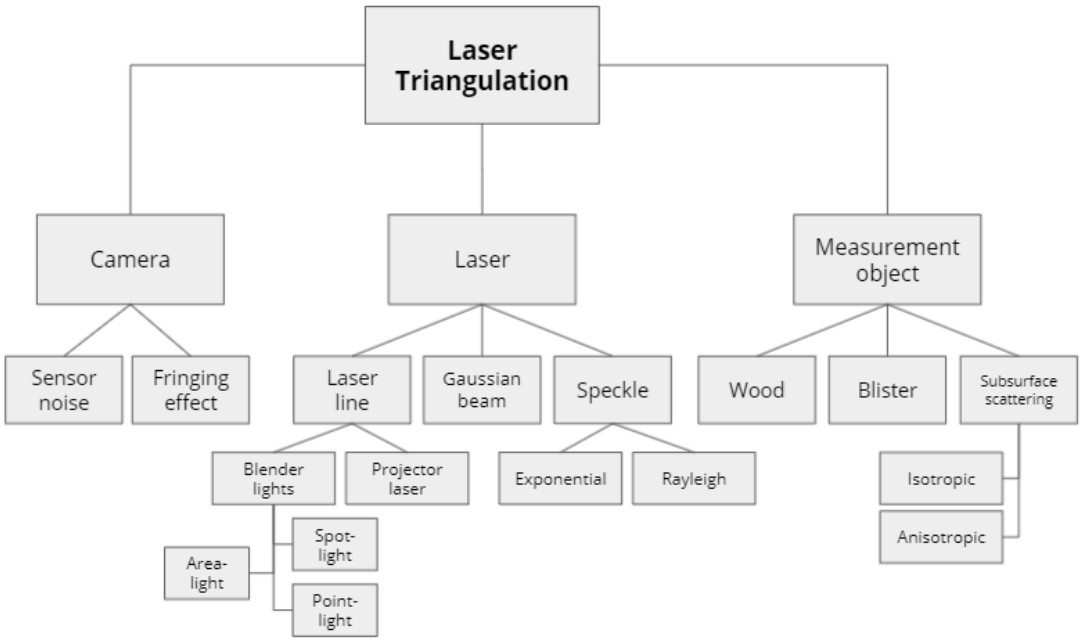


Figure 3.1: Overview of the implemented methods for simulating a laser triangulation system.

3.1. Practical laser triangulation

In order to create a simulation as realistic as possible, reference results were needed to enable comparison. These results were collected through practical laser triangulation sessions of two physical measurement objects. These objects consisted of a wooden plank and a partially emptied blister package. The plank was from a spruce, seen in *Figure 3.2*. The blister package used was from a pack of Ibuprofen Orifarm 400 mg film coated tablets, seen in *Figure 3.3*.

Figure 3.2: Raw wooden plank from a spruce, with three knots, two to the right and a small one far to the left. There is also rot along the entire top of the plank and some regions of dirt to the left. The fibre direction is along the horizontal axis of this image.

Figure 3.3: Blister package of Ibuprofen Orifarm 400 mg film coated tablets. The centre blister of the top row and the leftmost blister of the bottom row are empty.

To be able to achieve a trustworthy simulation, crucial properties and data were gathered from the instruments that were used in the physical laser triangulation (laser, sensor and camera). Optical power of the laser, various parameters of the camera and optics (focal length, aperture, and so on), as well as the resolution and noise properties of the sensor were needed to resemble the physical scene. Different parameters were used for the two measurement objects, due to their separate characteristics. The blister package reflected more light than the raw wood and therefore a shorter exposure time was programmed for the blister object.

SICK enabled the physical laser triangulation to be performed in an environment not affected by external lighting. This was achieved through the use of a selective wavelength filter

that blocked all wavelengths but the ones interesting for the triangulation. The laser triangulation set-up used by SICK to triangulate blisters and wooden planks worked by successively moving the measurement object (and not the laser) along one axis.

A red line laser, a 25 millimetre lens and camera with a CMOS-sensor was placed at 420 millimetres above the measurement objects. The laser had a wavelength of 660 nanometres ($\pm 15nm$) and a red band-pass filter was used to exclude other wavelengths. The camera was placed at a -25° angle from the baseline (axis perpendicular to the measurement object) and a Scheimpflug-adapter was used to get focus throughout the whole laser plane. The system was pre-calibrated and utilised global shutter technology. Sensor images or profiles were captured for every 6.8×10^{-5} m displacement of the measurement object. 4000 profiles were combined to create the resulting scattering images for the wooden plank, whereas 1000 profiles were used for the blister package.

3.2. Simulated laser triangulation

Several methods were tested for modelling each part of the laser triangulation system. Different plug-ins were considered and tested for the simulation of laser lines and sensors/scanners, such as BlenSor[42], Gazebo [43] and MORSE [44]. BlenSor is a simulation package specifically designed for sensor simulation. BlenSor was of interest since it can handle complex scenarios and analyse algorithms. Gazebo and MORSE are robot simulation plug-ins. Gazebo was interesting since it could generate sensor data with noise. Gazebo and MORSE also allowed for emulation of a line laser. MORSE was particularly interesting since it also offered a model of a laser scanner from SICK. Research showed that all three plug-ins were built for and/or assumed that the user ran Linux/Ubuntu [44][42][43]. The machines used during this thesis used Windows as operating system. Despite this, MORSE was installed together with the required old versions of Python and Blender. The other plug-ins were not further considered due to the results of testing MORSE. The installation and outdated documentation/requirements was considered too time consuming and not proportional to the outcome of testing a plug-in. More on this in the result section 4.2.

The simulation was performed without any external lighting, in accordance with Fernandez et al. [12]. Thus, both the simulated and the physical laser triangulation results were obtained in scenes only affected by the laser light. The movement of the simulated laser triangulation system was also executed in the same manner as in the physical system, by moving the measurement object along one axis between each frame. Sensor images were simulated by rendering 2D-images of a Blender scene including a laser line projected onto a measurement object. As the measurement object was moved, new frames were rendered, resulting in simulated sensor images.

3.2.1. Laser line simulation

Several techniques were adopted when simulating laser lines in Blender. Among them were two fast methods, along with a more complex and time consuming one. One quick laser simulation was performed by reshaping the different built-in light sources of Blender using two techniques. The first reshaping method was to use slits created by geometries, whereas the second approach was to utilise a tree of shader nodes. Another quick method was to transform the built-in *point-light* to a projector that projected a pre-constructed 2D image texture representing a laser. The more high-end laser simulation technique involved approximating a point laser as a Gaussian beam with a tree of shader nodes and then reshaping it using a cylinder lens. All methods had the common aim to resemble a realistic laser with varying thickness and intensity. Speckle was another laser property that was considered important for realism and therefore a phenomenon to be applied to all simulated lasers.

Reshaping Blender's built-in light sources

The built-in light sources considered were *point-light*, *spot-light* and *area-light*. Blender's *sun-light* was excluded due to its unsuitable property of being emitted from infinitely far away. As previously stated, two reshaping methods were tested. One of them was placing the light on one side of a small opening or slit, which was created by two closely placed geometries as seen in *Figure 3.4*. On the side of the slit that was opposite to the light source, the light was shaped into the form of the opening. Two planes with diffuse materials were created and positioned with a thin space between them. By placing these two planes between the light source and the object to be measured, the light source was reshaped into a line with the same width as the opening between the planes.

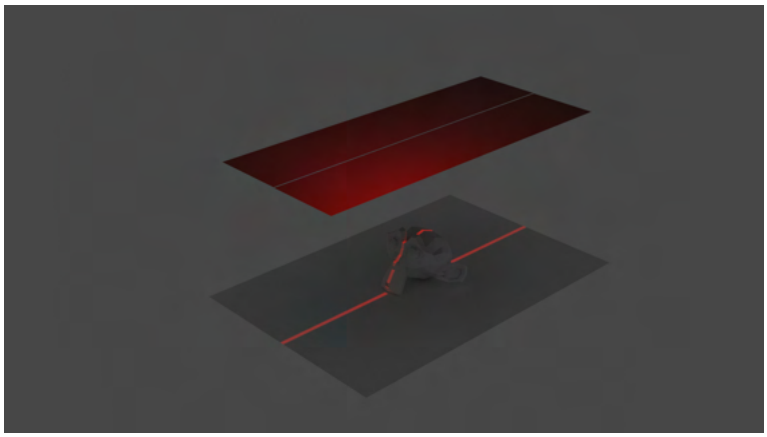


Figure 3.4: Laser line created using Blender's built-in point light, placed above two diffuse planes with a small opening between them.

An alternative approach for altering the shape of a light was to incorporate collections of shader nodes. Using basic vector maths through Blender's shader nodes, the incoming light ray at every texture coordinate was scaled along each axis into a beam travelling in the wanted direction, never thicker than a given width. The same node tree could be used for all light sources. The normals of light sources were approximated, with knowledge from the triangulation scene, for all lights but the area-light, which had an actual normal. The node tree for reshaping the area-light could therefore be simplified.

Projecting a laser texture

As another laser simulation, a texture projector was created with Blender's built-in point-light. A 2D image texture of a laser with Gaussian distribution was generated using a raster graphics editor (image editing software). A three pixels wide line was drawn and blurred using Gaussian blur with a five pixels radius, see *Figure 3.5*. The texture was applied as a non-repeatable image texture using the *Image texture* node. The texture was then mapped and scaled to resemble reference pictures, using different shader nodes.

Figure 3.5: Laser texture created in a raster graphics editor used to simulate a line laser with a point-light projector in Blender.

Gaussian beam approximation

A Gaussian beam was approximated as an emitting volume in Blender, through a tree of shader nodes and by following the theory presented by Bergmann et al. in their work from 2016 on integrating Gaussian beams into a physically-based renderer [9]. The radiance was computed for each texture point in the volume, considering the beam waist, power, wavelength and width of the laser beam. Typical laser behaviours were obtained mainly by mimicking the transverse Gaussian intensity profile and by varying the width of the beam depending on the lateral distance to the centre of the beam. The derived radiance was then used as input to the *strength*-parameter of the emission node. For more details on Gaussian beam theory, please refer to the source report [9].

To reshape this Gaussian point laser, a cylinder lens was constructed as a geometry with a glass-like material. Early findings showed that it is currently not possible to render sufficiently realistic caustics in Blender Cycles and therefore, another renderer was tested. LuxCoreRenderer was the new renderer of choice due to its alleged focus on physically based light simulations [29]. A working cylinder lens was given in one of the example scenes included in LuxCoreRenderer's documentation. The texture node tree of the emitting volume had to be translated into LuxCoreRenderer's material nodes, since not all Cycles-nodes were supported in the new renderer.

Speckle simulation

Speckle patterns were applied to simulated laser lines through a post-processing Python script, based on the methodology presented by Bergmann et al. [9]. A stack of QxQ zeroed matrices was generated as a base for the speckle patterns. Q was an arbitrary integer determining the size of the speckle pattern images. Further, a single QxQ noise matrix of random complex numbers was created. The complex numbers were of unit amplitude and their imaginary parts were uniformly distributed over the range $[0, 2\pi)$. A circle was cut out from this noise matrix and inserted into each of the zeroed matrices. The circles were cut out from and placed at positions with a constant radius from the matrix centre and at different angles. This entailed that no matrices had their circles placed at the same location, see the top row of *Figure 3.6* and the left illustration of *Figure 3.7*. A Fourier transform was applied to the matrices, followed by pixel-wise multiplication with the squared magnitude of the pixel. These operations resulted in a unique speckle pattern for each of the matrices in the original stack, see middle and bottom row of *Figure 3.6*. All patterns were correlated to each other, where the correlation depended on the amount of overlap of the circular masks from which the patterns were generated.

These patterns were then applied through pixel-wise multiplication with the simulated sensor images. The smallest speckle size of the generated pattern $(1.2\mu m)$ was smaller than the pixels of the sensor $(6\mu m)$, giving a sample factor as $\frac{6}{1.2} = 5$. Therefore, each sensor image pixel (x, y) was computed as the average of the corresponding $5x5$ grid of pixels in the speckle pattern. The speckle pattern coordinates of the grid (x_p, y_p) were wrapped using the modulus operator to stay within the range $[0, Q-1]$, see equation 3.1.

$$\begin{aligned} x_p &= x \mod (Q-1) \\ y_p &= y \mod (Q-1) \end{aligned} \tag{3.1}$$

Any displacement in the laser triangulation system should cause the speckle pattern to be updated. This was done using the varying correlations of the patterns. Each system translation was represented by a specific shift of the angle of the complex numbered circle, meaning that each movement corresponded to a specific speckle pattern. If the movement matched an angle between two patterns, linear interpolation was performed. To determine which pattern(s) from the stack to apply pixels from, a third dimension (u) was added to the speckle pattern coordinate, see *Figure 3.7*. Several simplifications were made to the computation of

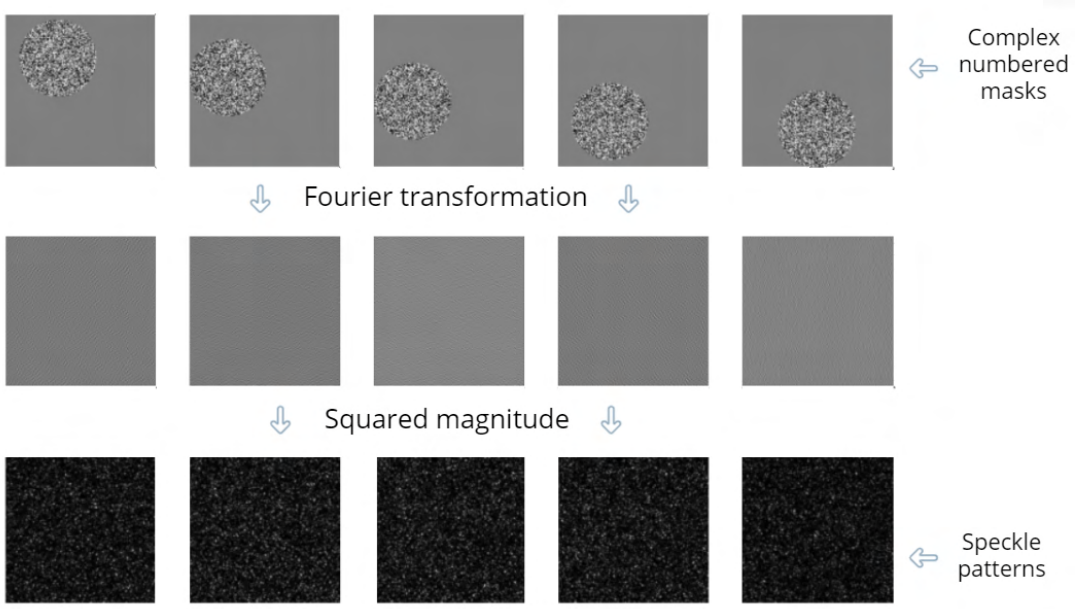


Figure 3.6: Illustration of speckle pattern construction, in accordance with the method presented by Bergmann et al. in their work *A Phenomenological Approach to Integrating Gaussian Beam Properties and Speckle into a Physically-Based Renderer* from 2016 [9]. The top row showing matrices with a circular mask of complex numbers, the middle row showing the Fourier transform of these masks and the bottom row illustrating the final speckle patterns.

u. This was possible due to the fact that the simulated laser triangulation system only included movement in one dimension; the distances and angles between every system part were kept constant. The displacement of the measurement object was of the same size between each profile, allowing for further simplifications. The simplifications for calculating u are presented in Appendix A. Refer to the source report [9] for more detailed information on this method.

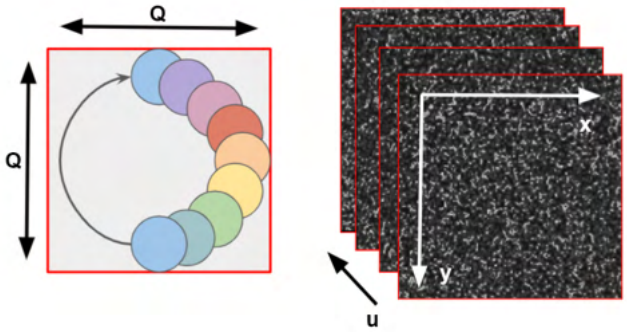


Figure 3.7: Illustration of speckle pattern correlations and coordinates, in accordance with the method presented by Bergmann et al. [9]. The correlation between patterns corresponds to the amount of overlap of the QxQ circular masks from which the patterns were created (left). A third coordinate u used to determine which pattern(s) from the stack to consider (right).

In accordance with Bergmann et al., the laser was assumed to be perfectly coherent and polarised and the speckle patterns were implemented with exponential intensity distribution [9]. In the limit of zero degree of polarisation, Duncan and Kirkpatrick state that the inten-

sity distribution of a speckle pattern is a Rayleigh distribution. Applying the square root to the squared magnitude of the Fourier transformed masks would reshape the exponential distribution to the wanted Rayleigh one [30]. Even though lasers are polarised with higher degrees of polarisation, speckle patterns with Rayleigh distribution were implemented. This was done partly to enable comparison of patterns with the two intensity distributions, but also to better resemble ground truth images.

For both distributions, energy was lost when the speckle pattern was applied to the laser line. This was due to the fact that the majority of the QxQ matrices were zeroed after insertion of the circle as well as after the Fourier transform. A simple scaling factor was therefore added to ensure that the algorithm was approximately energy preserving. The intensity of the speckle pattern was then scaled further to better resemble the different exposures of the reference images. An overview of the speckle pattern simulation is seen in the following list:

1. Generate a stack of QxQ zeroed matrices.
2. Create a QxQ noise matrix of random complex numbers.
3. Cut out circles from the noise matrix (at same radius but different angles) and insert one into each zeroed matrix (at the same position as they were cut from).
4. Apply a Fourier transform to each circular mask (zeroed matrix with inserted circle).
5. Multiply the masks from the previous step pixel-wise with the squared magnitude of the pixel.
6. Change the intensity distribution from exponential to Rayleigh by taking the square root of the masks from the previous step.
7. Scale the intensity distribution to approximate conservation of energy.
8. Compute the $sampling\_factor$ as sensor pixel size divided by smallest speckle size.
9. For each sensor image pixel, derive pattern coordinates for a $sampling\_factor \times sampling\_factor$ grid. Wrap the coordinates: $(x, y) \in [0, Q-1]$ and $u \in [0, \#patterns]$
10. Multiply the sensor image pixel-wise with the average of the corresponding grid.

3.2.2. Camera simulation

The built-in camera in Blender allowed the following parameters to be altered: aperture, focal length, focus distance and sensor size. Data from the physical laser triangulation scene was used to resemble the used camera in accordance with *Table 3.1*. The physically measured distance between the object and the camera was also recreated in the scene. The simulated camera was assumed to have a perfect and aberration-free lens and the Scheimpflug-adapter that was used in the physical laser triangulation, was completely neglected in the simulation.

Table 3.1: Camera parameter values in Blender for simulating a laser triangulation system

Camera parameter	**Value**
Aperture (f-stop):	2.8
Focal length:	25 mm
Focus distance:	∞
Sensor size (width):	15.36 mm
Distance camera-object:	410 mm

Sensor simulation

As discussed above, Blender allowed input parameters for sensor size, but to simulate the noise of the sensor, a post-processing Python-script was implemented. A ground truth sensor image from a physical laser triangulation session was used to plot histograms of the noise distribution. To identify the intensity distribution profile of the noise, only regions with no (or very little) laser impact were considered. The profile was identified as approximately Gaussian and the mean and standard deviation of the ground truth sensor image was extracted. These parameters were then used to create a matrix of random numbers with Gaussian distribution, which was later added to the simulated sensor images.

Fringing effect

Looking at ground truth images, it was seen that some unknown phenomenon occurred around the laser line making it look "hairy", see *Figure 3.8*. This effect could possibly be blooming, a material effect, pixel bleeding or a combination of several events. This effect was simulated using Gaussian blur as a simplification.

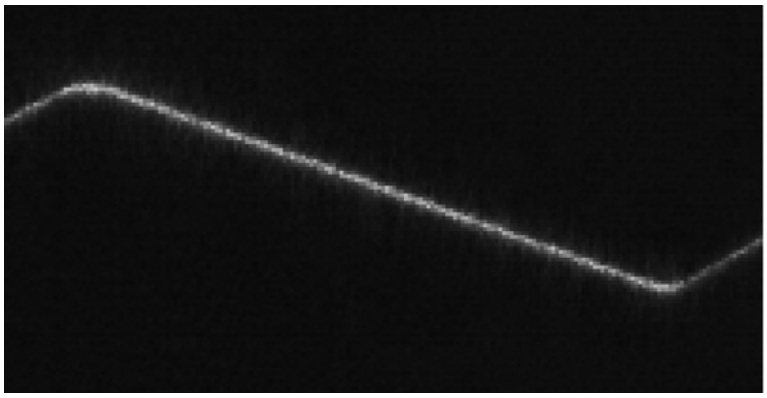


Figure 3.8: Magnified region of ground truth sensor image showing an unknown bleeding/blurring/blooming effect around the laser line.

3.2.3. Measurement objects

The objects to be measured were simulated as geometries with associated materials in Blender. Tracheids (grain direction) and branches (knots) were considered the most important features of the wood, whereas the leading property of a blister package was the difference in the pill and package materials.

Raw wooden plank

Measurements of the reference plank were used to create the geometry for the simulated wooden plank. A primitive cube was modelled into a plank of the following dimensions: 94x210x22 (mm).

Several approaches were tested for creating a spruce wood material with fibre directions and knots. The first method was to create a volumetric 3D-material from 2D-texture images. Microscopic photographs of the three different wood planes (transverse, tangential and radial) were loaded into Blender with the *Image texture* node. The textures were then mapped, scaled and repeated onto a geometry, so that the level of detail and density resembled real wood. This volumetric texture was applied alone as well as in combination with various surface materials.

Another approach to simulate wood was as a heterogeneous volume with LuxCoreRenderer. LuxCoreRenderer supported creating complex volumes through the *Heterogeneous Volume* texture node [45]. Several node trees were constructed and applied to plank-like geometries.

Yet another method was to generate a procedural texture using Blenderkit [46]. Blenderkit is an add-on for Blender which offers free textures. The chosen procedural texture had high level of detail with adjustable parameters for lines and knots to resemble spruce wood.

To have more freedom and understanding with the procedural texture, a custom procedural texture was created. This was done by combining the existing nodes of Blender to replicate desired properties of a wooden plank.

Furthermore, another method was to include a photograph of the wooden plank that was used in the physical laser triangulation system, seen in *Figure 3.2*. This image was then cropped, rotated, scaled and UV-mapped onto the wood geometry in Blender.

Subsurface scattering

There are several ways of including subsurface scattering in Blender materials. The *Principled BSDF* shader node, for example, has input fields for *Subsurface Radius* for each of the RGB-channels as well as the parameter *Subsurface*, which is used in the mixing of diffuse and subsurface scattering as a scaling factor for the scattering radii. For more control of the scattering, there also exists the *Subsurface Scattering* node (as well as the *Volume scatter* node), specifically made for handling this phenomenon [47]. However, none of the shader nodes that include subsurface scattering can be used to simulate 360° anisotropic (directional) subsurface scattering. The *anisotropy* parameters in the various shader nodes have shown through experiments to only be able to produce uniform forward and backwards scattering. Since raw wood experience anisotropic scattering, an external Python-script was implemented to simulate anisotropic subsurface scattering.

Anisotropic subsurface scattering simulation

In order to simulate directional scattering, a method for representing different scattering directions was needed. In an image describing a measurement object, each pixel should be assigned a direction value corresponding to the direction in which incoming light should scatter the most. This was done by mapping the scatter direction of each pixel as a vector with direction and unit magnitude. To, as effortlessly as possible, generate images to describe scattering directions of an object, directions were represented as RGB-colours. This enabled easy production of direction maps with areas of different colours. These coloured direction maps were created by interpolating between different colours using a gradient tool.

The anisotropic subsurface scattering script was built to scatter light along a given slope in both directions (positive and negative). Due to this, a half circle was enough to represent all the possible slopes. Colours between the red and green axes corresponded to slopes in the first and third quadrants of the unit circle, given by $\frac{R}{G}$. Similarly, slopes in the second and forth quadrants were represented by colours between the blue and green axes through $-(\frac{B}{G})$.

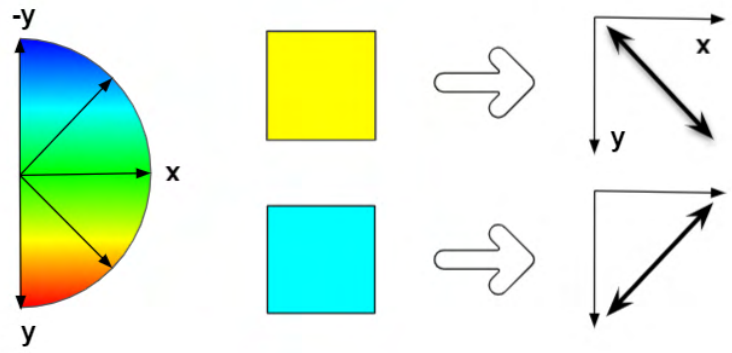


Figure 3.9: Illustration of how to represent directions as RGB-values. A yellow colour represents scattering in the ascending diagonal and a cyan colour corresponds to scattering in the descending diagonal.

Figure 3.9 roughly illustrates how the various colours were mapped to represent different directions. As an example, a yellow direction map (R = G = 255) corresponds to subsurface scattering along the ascending diagonal. The alpha channel of the colour was utilised to represent the scatter intensity, which described how much of the incoming light that each pixel should scatter forward in a given direction. In areas with lower alpha-values, it would scatter less, meaning that the pixel intensity would be spread a shorter distance. The coloured direction maps were then translated into a normalised vector field as seen in *Figure 3.10*.

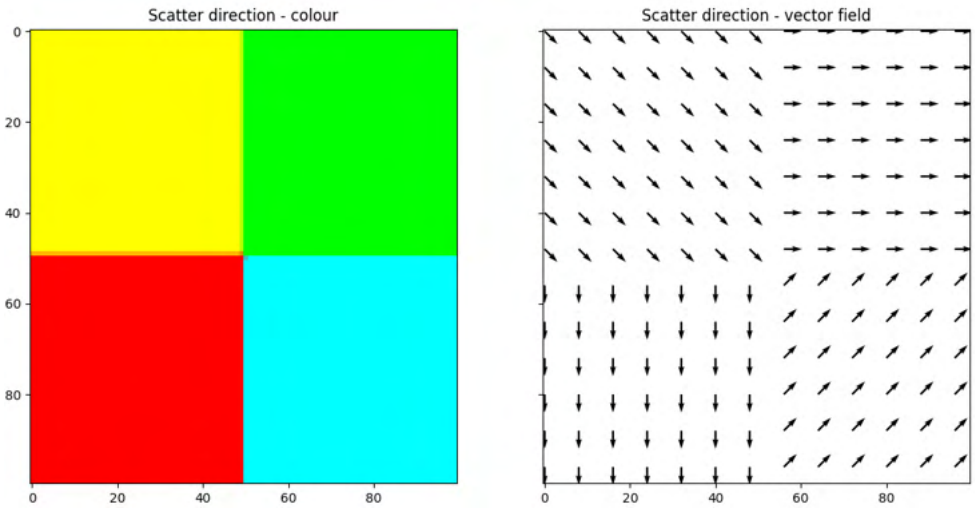


Figure 3.10: Illustration of how a subsurface scattering direction map is translated from RGB-colours to a vector field.

Once the scattering directions had been generated for an object, the subsurface scattering could be simulated as an iterative process, scattering more light for each iteration. The two photographs seen in *Figure 2.7* were used as references for the simulation. One illustrated scattering along the tracheids and the other showed scattering of a laser point placed on a knot. One zeroed matrix was created for each of the two reference images. Two laser points were then simulated as circles with diameter of 55 pixels and with Gaussian distribution. One laser point, placed on top of the knot of the wooden plank, was inserted into one of the matrices, whereas the second laser point was placed beside the knot and inserted into the other zeroed matrix.

Different approaches were tested for the scattering simulation, where the first one was to loop over each matrix and find the pixels that were lit by the laser. From these pixels, the intensity was spread into a number of nearby pixels in the direction retrieved from the direction map. Several methods were implemented for finding which pixels to scatter to, following the given direction. The initial method was to calculate the linear equation $y = kx + m$ for each direction and lit pixel. The coordinates of the pixels to scatter into could then be computed using the lit pixel's coordinate incremented along one axis. Bresenham's line algorithm [48] was also implemented, as well as a third line algorithm utilising the Python library *Scikit-image*. From the module *Draw* of this library, a function for generating pixel coordinates of *anti-aliased* lines was tested.

An alternative method for simulating anisotropic subsurface scattering included implementing the first part of a stable fluid solver as a diffusion process. Only the first part of the method was considered since it was based on a static vector field, rather than the dynamic ones used later in the algorithm. This method was performed in accordance with the paper published by Jos Stam [49], with an extension implemented to allow anisotropic scattering.

Stam's diffusion process was an iterative method for evenly spreading pixel values whilst obtaining a stable system. The method was based on the Navier-Stokes equations together with Gauss-Seidel relaxation. Stam's algorithm used the same coefficients for the four neighbouring pixels (above, below, to the left and to the right) to obtain a uniform spread and a stable flow [49]. By adding the four diagonal pixels, see *Figure 3.11*, the flow was distributed over more directions in each iteration. The intensity spread was then directed by having the coefficients for each neighbouring pixel depend on the directions of a given vector field. Stam's uniform flow was thus transformed into a simulation of anisotropic scattering.

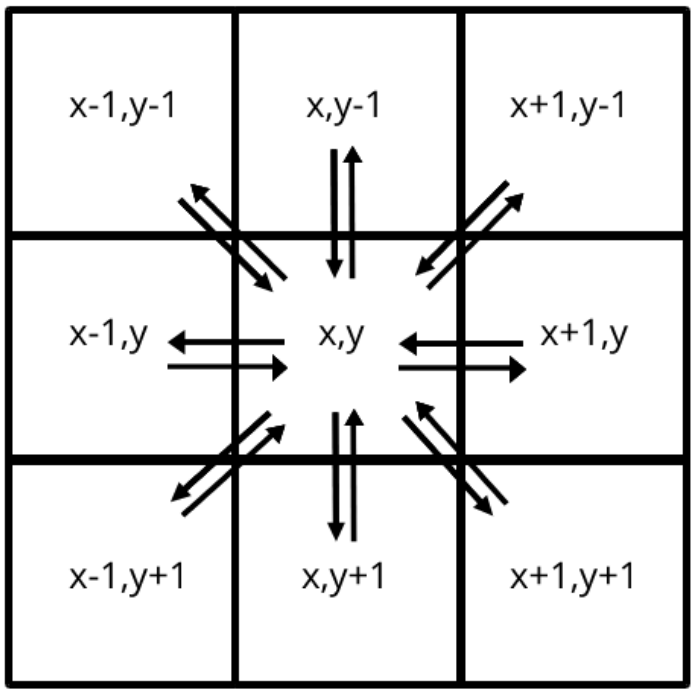


Figure 3.11: Illustration of uniform diffusion between neighbouring pixels, with the diagonals as an extension to the methodology presented by Jos Stam in his work *Real-Time Fluid Dynamics for Games* from 2003.

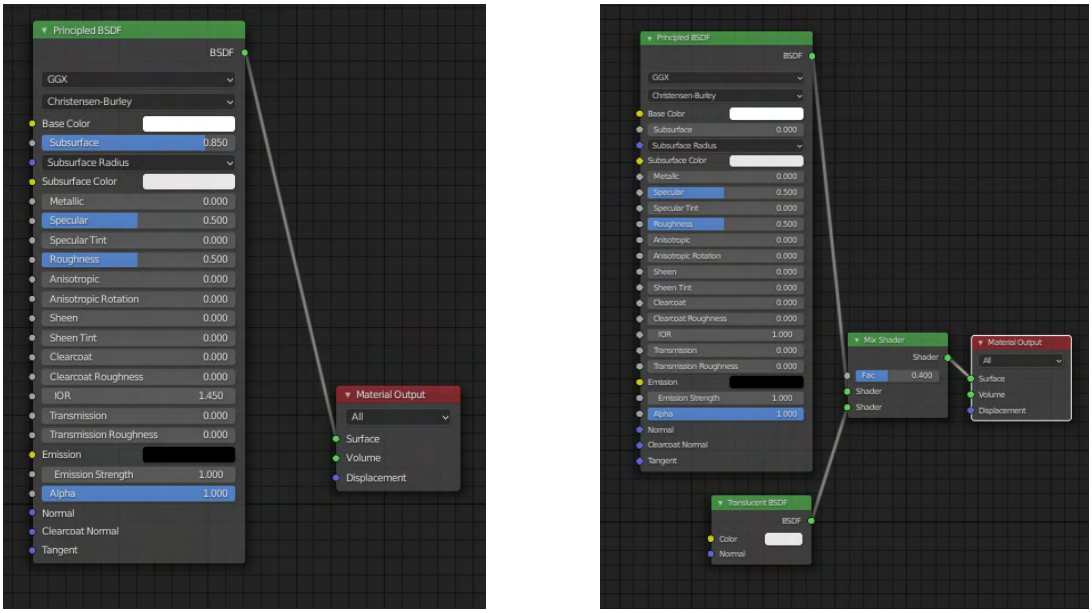
In order to simulate the scattering in wooden knots, the mentioned alpha-values were used. Since the fibres of the knots were perpendicular to the tracheids, as presented in section 2.3.4, the light transportation could be simulated as absorption. The lower the alpha-value of a pixel, the more absorption and the less scattering would occur in that pixel. Measurements from the practical laser triangulation session showed that the intensity of the reflected light was about half as high in the knot as in the clear wood. Suitable alpha-values in the knot-regions of the coloured direction map were therefore set to 0.5. The scattering in these knot-areas was set to be uniform, to resemble the reference image, by giving all the diffusion direction coefficients the same value. A lower number of diffusion iteration was used for the laser point placed on the knot than for the one placed on the clear wood. This was also done to achieve scattering effects that were similar to the reference images.

Absorption was performed in a separate pass where the whole image was multiplied pixel-wise with the corresponding alpha-value. For the diffusion approach, the absorption algorithm was applied between each diffusion iteration. The alpha-values were therefore divided by the number of iterations so that the combined effect of all the absorption iterations corresponded to the given alpha-value for each pixel. Due to these absorption passes, the diffusion process was no longer energy conserving. An overview of the algorithm for simulating anisotropic subsurface scattering is seen in the following list:

1. Describe the scattering directions and absorption of different areas of a photograph using RGBA-colours.
2. Transform the colour of each pixel in the map from step 1 into a slope of a vector to generate a vector field.
3. Create a circle with Gaussian distribution to simulate a point laser.
4. Diffuse or iteratively spread the intensity of the laser point following the directions of the vector field, for a given number of iterations.
5. Scale the spread intensity using the alpha-value of each pixel to simulate absorption.

Blister package

The blister package geometry was created by merging half ellipsoids with a plane. Some of the blisters contained pills that were shaped as whole ellipsoids. The reference blister package consisted of PVC and aluminium foil, but since PVC was the main component of the package and also the material where the laser hit, the simulation was performed with a single material for the entire package. Two Blender materials were constructed, one for the blister package and one for the pills. The package material was created as a mix between the *Translucent BSDF* shader and the *Principled BSDF* shader with white as base colour and with the *Subsurface* parameter set to zero. The pill material was also white, but fully opaque and therefore simulated using only the *Principled BSDF* shader, with a *Subsurface* coefficient just below one and *Subsurface Radius* of one for each colour channel. Screenshots of the two materials are seen in *Figure 3.12*.



(a) (b)

Figure 3.12: Shader node trees for simulating a blister package in Blender, showing the tree of a pill material (a), as well as a tree of a blister package material in (b).

3.3. Post-processing

2D-sensor images were rendered from a full laser triangulation scene, including a measurement object, a camera, and a laser. These simulated sensor images were then sent through the three previously presented post-processing scripts to apply Gaussian blur, speckle pattern and simulated sensor noise.

For the sensor image comparisons a Blender 3D-model of a saw-tooth object was given by SICK. This measurement object was SICK's main piece for calibrating laser triangulation systems and can be seen in *Figure 3.13*. Therefore, a realistic model was created and a large amount of ground truth images was available for this object. The real sensor images of the saw-tooth also included and showed the interesting artefact of speckle patterns clearly, making them suitable for evaluating the realism of the simulated speckle patterns.

Figure 3.13: Photograph of SICK's calibration object RCAL-300, which is a 300 millimetres long saw-tooth with 15 teeth [50].

The given saw-tooth model was included into the Blender scene and sensor images were simulated with two types of lasers: the node-reshaped point-light laser and the texture projector laser. Several images were created in the post-processing of each sensor images, where

different numbers of scripts were applied in different orders. Following are examples of a few post-processing scripting orders (leftmost script applied first):

- sensor noise, speckle
- Gaussian blur, sensor noise
- speckle, Gaussian blur, sensor noise
- Gaussian blur, speckle, sensor noise

The parameters of each script were also altered to resemble the ground truth images as much as possible. Some of these variables were the radius of the Gaussian blur and the size of the smallest speckle in the speckle patterns. Final parameter values are seen in *Table 3.2* below.

Table 3.2: Post-processing script parameters for each measurement object and for the three scripts blur, speckle and noise.

Script parameter	**Saw-tooth**	**Wood**	**Blister pack**
Blurring radius:	0.5	1.0	0.5
Speckle pattern size: (pixels)	1000x1000	1000x1000	1000x1000
Mask diameter: (pixels)	500	500	500
Intensity scale factor:	3.5	6.0	7.0
Sampling factor:	5	5	5
Mean:	11.6	11.6	11.6
Standard deviation:	1.09	1.09	1.09

3.3.1. Simulating scattering images

Simulated wood and blister packages were used to simulate the subsurface scattering images. 1000 profiles were rendered of the scene with a blister package and 4000 profiles were created for the wooden plank scene. These simulated sensor images were all lit by the same texture projector laser and both of the measurement objects were moved 6.8×10^{-5} m along the y-axis between each profile. All three of the post-processing scripts were applied to each sensor image, with different parameters for the two objects, see *Table 3.2*.

SICK provided an algorithm for combining the profiles into a subsurface scattering image. Laser line extraction was included in this algorithm and performed before the subsurface scattering was measured. The laser centre was derived through centre of gravity. From this detected laser line, the area in which to compute scattering was defined by the two parameters *scatter offset* and *scatter range*. *Scatter offset* was the distance (in pixels) from the detected laser line to the first pixel to be measured. From this pixel the *scatter range* defined the width of how many pixels that scattering was to be measured in, see *Figure 3.14*. The parameter values for wood and blister packages are seen in *Table 3.3* below. All algorithms and inputs were set to match the settings of the physical laser triangulation sessions.

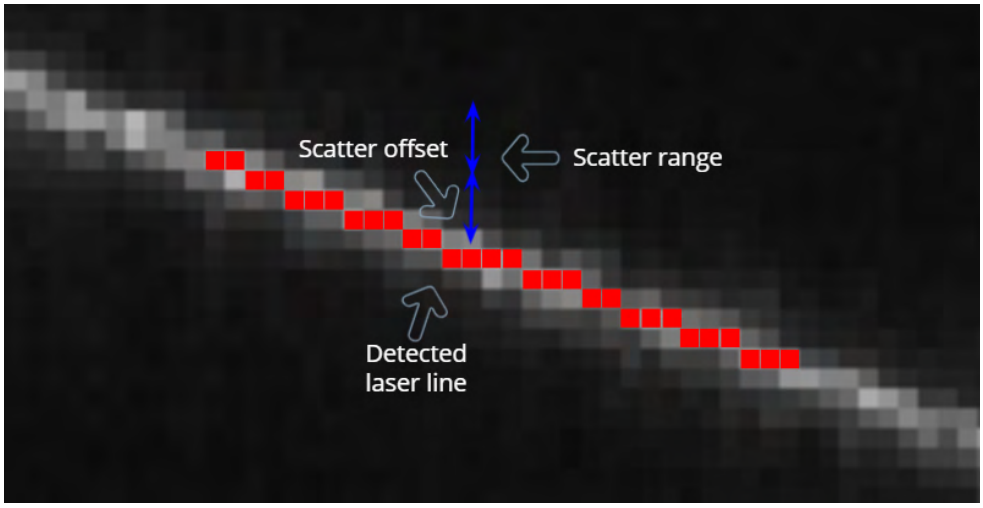


Figure 3.14: Illustration of the two parameters *scatter offset* and *scatter range* used in the algorithm to measure subsurface scattering in sensor images.

Table 3.3: Script parameters of the subsurface scattering measurement algorithm for the two measurement objects wood and blister package.

Script parameter	**Wood**	**Blister pack**
Scatter offset:	1.0	7.0
Scatter range:	8.0	5.0

3.4. Comparison and evaluation

The results from the simulated laser triangulation were compared to results from the practical laser triangulation. As presented, both triangulations were performed with similar set-ups and measurement objects. The outputs that were possible to compare were the sensor images and the subsurface scattering images.

The definition of a realistic line laser was considered as a laser that includes the main properties: intensity variation and visible speckle pattern. A realistic line laser would also visually resemble ground truth data. A material (for a measurement object) was defined as realistic, in this thesis, if it affected light in accordance with behaviours seen in ground truth images. The material should also match the visual appearance of the reference object.

4 Results

4.1. Practical laser triangulation

The practical laser triangulation sessions, covered in section 3.1, resulted in sensor and scattering images for both of the measurement objects: wooden plank and blister package. The sensor images of the blister package were overexposed, see *Figure 4.1*, destroying most interesting phenomena. Relatively small amounts of speckle were seen in the wood sensor images, making them unsuitable for comparison with the results of the simulated speckle patterns, see *Figure 4.2*.

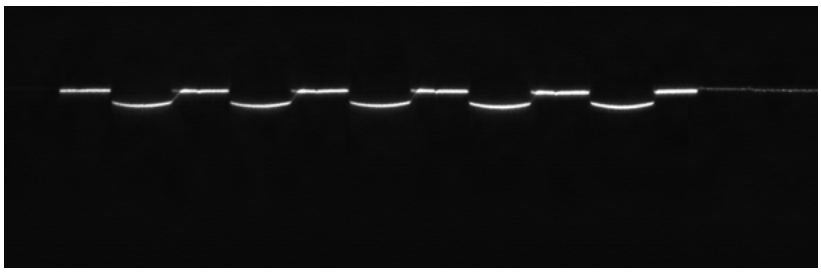


Figure 4.1: Result from practical laser triangulation session of a blister package, showing a sensor image containing a laser line projected onto the blister package seen in *Figure 3.3*.

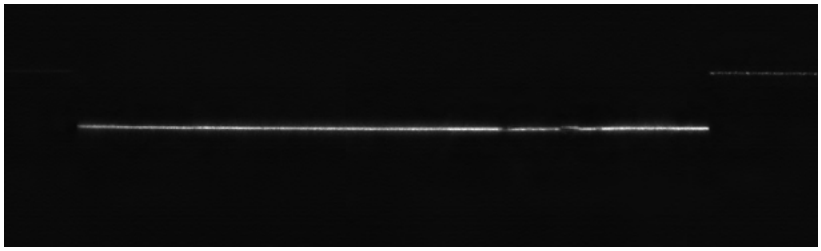


Figure 4.2: Result from practical laser triangulation session of a wooden plank, showing a sensor image containing a laser line projected onto the plank illustrated in *Figure 3.2*.

Results from a laser triangulation session, performed by SICK, of the previously mentioned saw-tooth object will therefore be used to compare the different laser line simulation methods as well as the post-processing scripts: sensor noise simulation, speckle pattern simulation and fringing effect simulation. A sensor image from the physical laser triangulation session of the saw-tooth is seen in *Figure 4.3*.

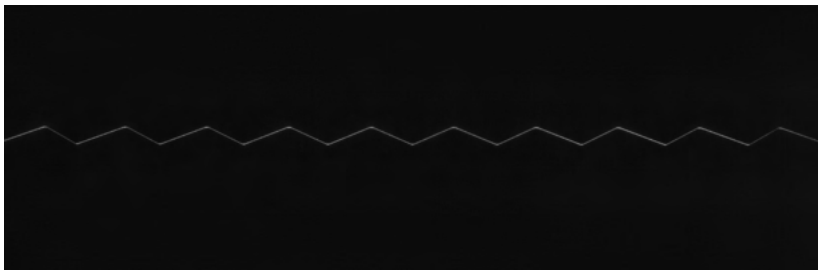


Figure 4.3: Sensor image from a practical laser triangulation session, performed by SICK, of the saw-tooth calibration object depicted in *Figure 3.13*.

4.2. Simulated laser triangulation

Three plug-ins were considered as an alternative method for simulating lasers and or laser sensors, as stated in section 3.2: MORSE, BlenSor and Gazebo. Installing MORSE resulted in no working lasers or sensors and it was decided that no plug-ins were to be used. The complexity in set-up along with the requirements for old programs did not seem proportional to the actual outcome of the plug-ins. Since BlenSor and Gazebo were also aimed for other operating systems than the one used in the project, and since they had similar requirements to MORSE, the idea of using plug-ins was cancelled as a whole.

4.2.1. Laser line simulation

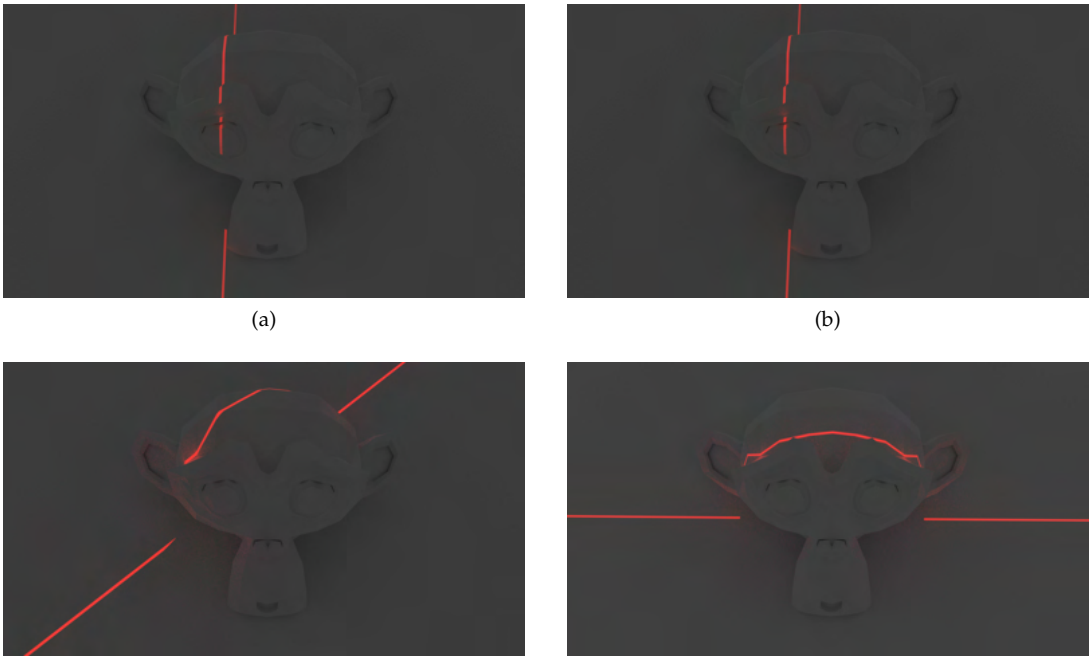
Laser lines were simulated, as presented in section 3.2.1, by altering the built-in light sources of Blender and by creating custom light sources as emitting volumes. Different approaches were used to reshape the different lights into laser lines. All simulated laser lines were projected onto the standard Blender geometry Suzanne and rendered with the same scene parameters.

Reshaping Blender's built-in light sources

Altering the shape of the built-in lights into laser beams was accomplished in Blender using shader nodes or a slit between two geometries, in accordance with section 3.2.1. Realistic properties such as varying thickness and intensity were however not achieved using these reshaping methods. *Figure 4.4* below shows a comparison of the two reshaping methods for a point-light. The slit and shader node methods resulted in identical laser lines with constant intensity and thickness. The results of reshaping the spot-light, point-light and area-light with shader nodes are seen in *Figure 4.5a-c*. No visual difference is seen in these rendered images, regarding the interesting factors of line shape or intensity.

(a) (b)

Figure 4.4: Results from laser line simulation in Blender Cycles presented in section 3.2.1, showing a point-light reshaped with shader nodes (a) compared to the same point-light reshaped with an opening between two planes in (b).



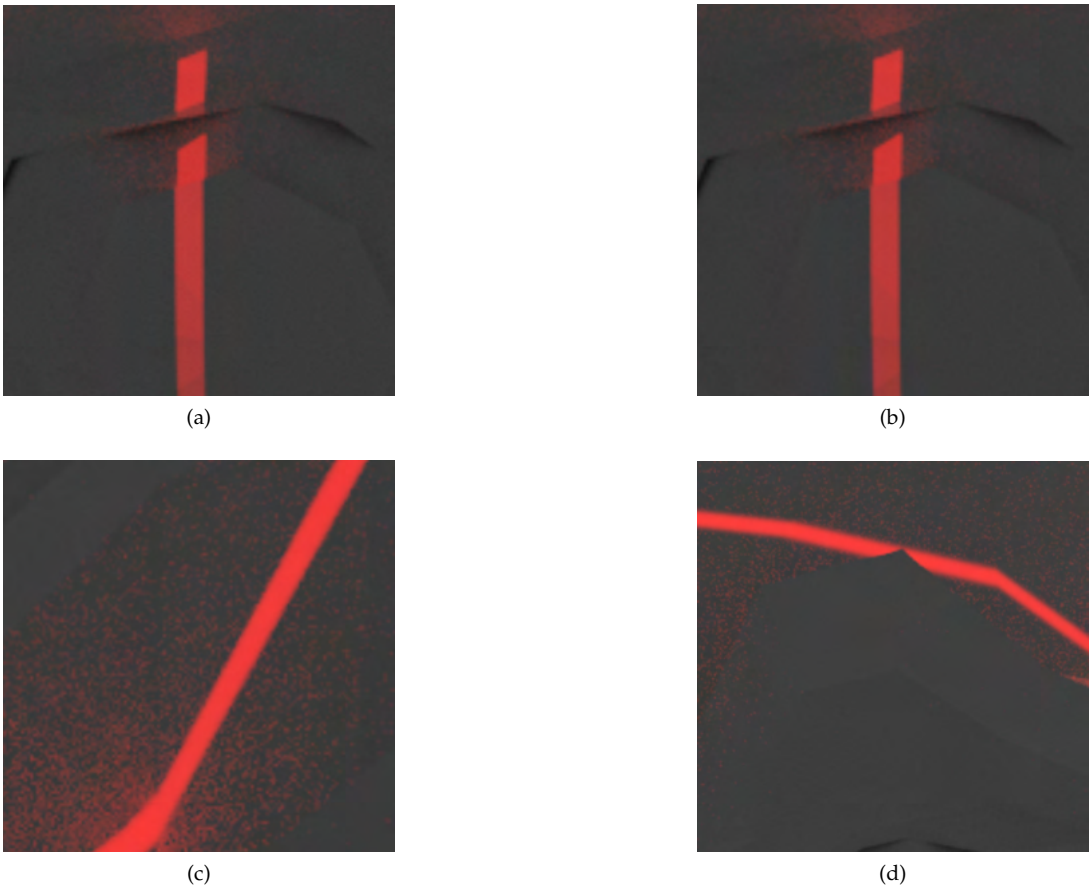
(a) (b)

(c) (d)

Figure 4.5: Results from laser line simulation in Blender Cycles, described in section 3.2.1, showing a reshaped spot-light in (a), a reshaped point-light in (b), a reshaped area-light in (c) compared to a point-light projector in (d). All light sources were altered using Blender shader nodes and the image texture used for the projector is seen in *Figure 3.5* in the method chapter.

Projecting a laser texture

The laser that was created by projecting the laser texture, as presented in section 3.2.1, resulted in the rendered image seen in *Figure 4.5d*. The four rendered laser lines in *Figure 4.5* are very similar in intensity and shape. Close-ups on these results are shown in *Figure 4.6* below, where the spot-light (*a*), point-light (*b*) and area-light (*c*) is seen to have sharper edges than the texture projector in (*d*).



(a) (b)

(c) (d)

Figure 4.6: Zoomed in results from laser line simulation in Blender Cycles, comparing the reshaped spot-light in (a), the reshaped point-light in (b), the reshaped area-light in (c) and the point-light projector in (d). All light sources were altered using Blender shader nodes and the image texture used for the projector is seen in *Figure 3.5* in the method chapter.

Gaussian beam approximation

Figure 4.7 depicts the Gaussian beam that was implemented as an emitting volume in Blender Cycles, as presented in section 3.2.1. The intensity of the laser varies along the transverse plane (cross section), from strongest in the centre, to weakest at the edges. The laser intensity is also gradually weaker further away from the beam waist.

Figure 4.7: Rendered image of a point laser simulated as a Gaussian beam, in accordance with the methodology presented in section 3.2.1 and with the desired laser properties of varying thickness and a Gaussian intensity distribution along the transverse plane.

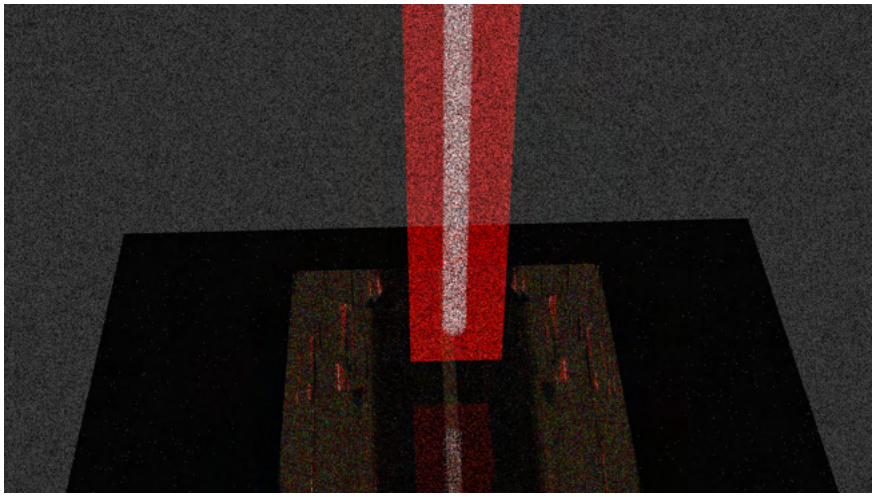


Figure 4.8: Results of the Gaussian beam initially implemented in Blender Cycles, see *Figure 4.7*, translated into the LuxCoreRenderer, as described in section 3.2.1.

(a) (b)

Figure 4.9: A scene where the built-in area-light laser of LuxCoreRenderer is reshaped into a laser line using a cylinder lens (a), together with a rendered image of the constructed laser line in (b).

This point laser was, as stated in section 3.2.1, not possible to reshape into a line using Cycles. An attempt to translate the Cycles shader nodes into LuxCoreRenderer material nodes resulted in the rendered image depicted in *Figure 4.8*.

The scene for LuxCoreRenderer also includes a cylinder lens, with purpose to reshape the point laser into a line laser. *Figure 4.9* shows a scene where the built-in area-light laser (incident from the right) is successfully reshaped into a line using a cylinder lens in *a* and a rendered image of the scene in *b*.

Speckle simulation

Stacks of speckle patterns were simulated with exponential and Rayleigh intensity distributions respectively, as stated in section 3.2.1. *Figure 4.10* includes one pattern of each distribution and illustrates that the exponential patterns were darker than the Rayleigh-distributed ones.

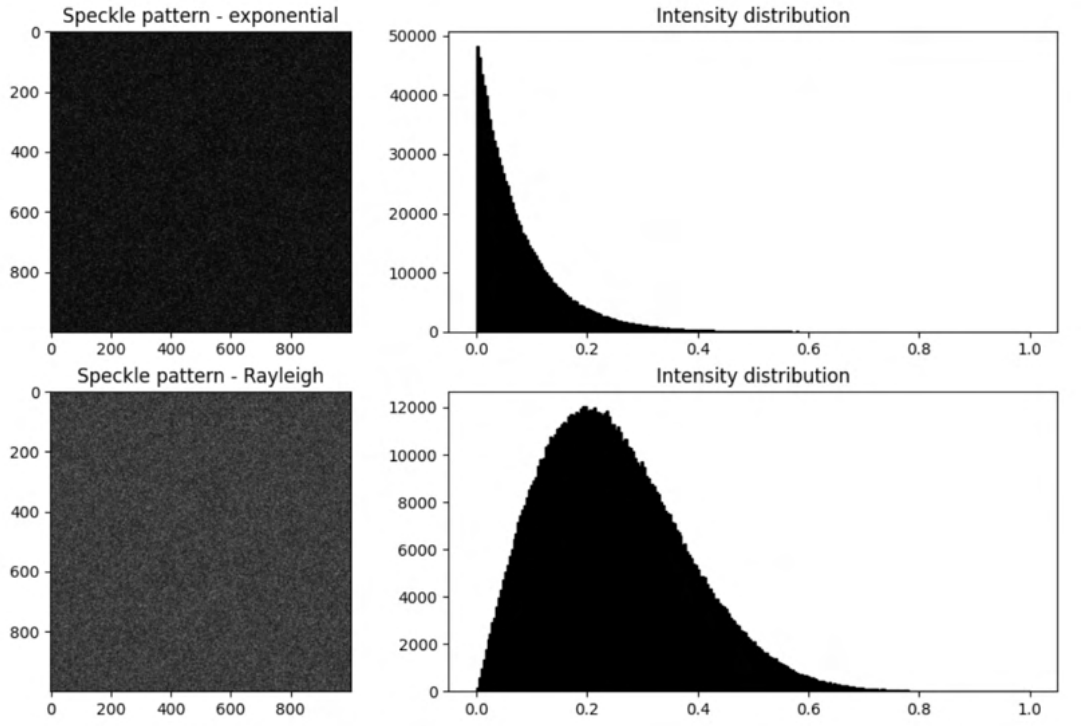


Figure 4.10: Results from the speckle pattern simulation presented in section 3.2.1, illustrating a pattern with exponential intensity distribution (top row), compared to a Rayleigh distributed speckle pattern (bottom row), with a corresponding histogram.

Figure 4.11 below depicts an attempt to match the intensity of the simulated laser lines that included speckle patterns (middle and bottom row) to the intensity of ground truth data (top row). The exponential speckle patterns (bottom row) were scaled with a factor of 10.0, whereas the Rayleigh-distributed speckle patterns (middle row) were scaled with 3.5. Simulated sensor noise was applied to both of the simulated sensor images. This resulted in laser lines with sufficiently small visual differences to the reference image. The simulated laser lines used for these comparisons were from the texture projector.

Figure 4.12 presents the simulated and Rayleigh-distributed speckle patterns applied onto rendered sensor images of the saw-tooth. The sensor images were created using either a reshaped point-light laser or a texture projector laser.

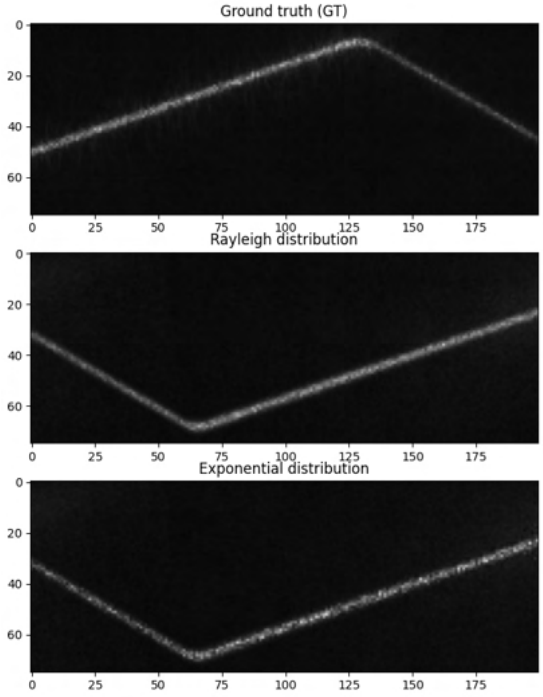
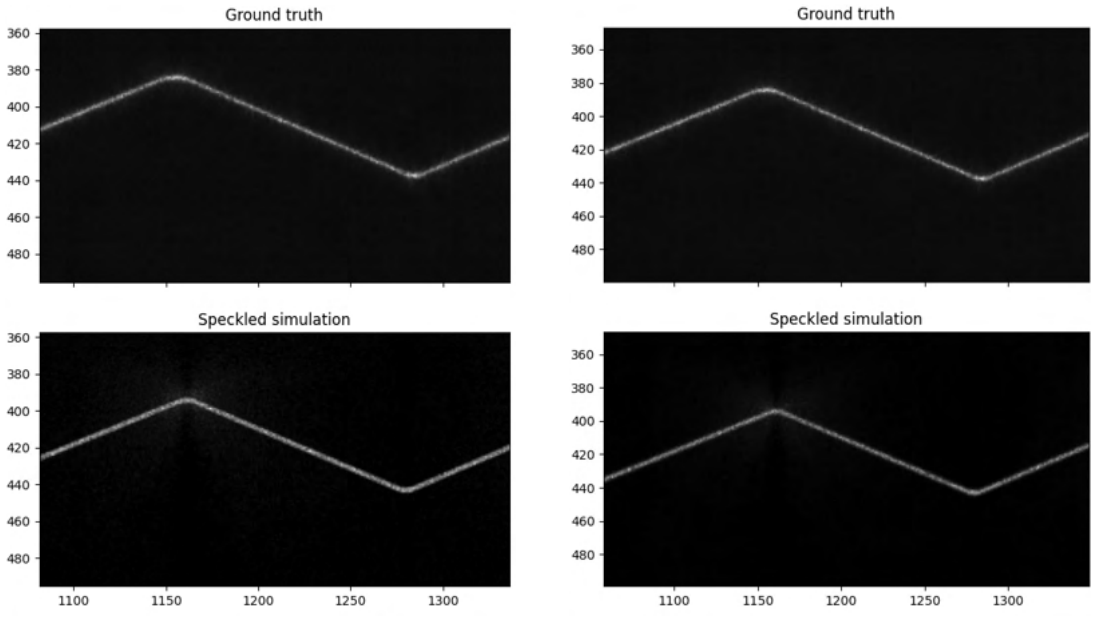


Figure 4.11: Results from attempts to match the visual appearance of ground truth speckle by applying scaling factors to the simulated patterns, as stated in section 3.2.1. The simulated speckle patterns were applied to sensor images rendered in Blender. Images of the laser lines including speckle are shown for the ground truth speckle (top row), the simulated Rayleigh-distributed speckle pattern (middle row) as well as for the exponential speckle (bottom row). Simulated sensor noise had been added to both of the simulated sensor images.



(a) (b)

Figure 4.12: Results from the speckle pattern simulation, described in section 3.2.1, showing the patterns applied to a reshaped point-light laser in (a) and to a texture projector laser in (b). Both results are compared to ground truth data from a practical laser triangulation session and the same values for the script parameters were used for the two different laser simulation methods.

4.2.2. Camera simulation

Inputting the camera parameters from the physical laser triangulation sessions into Blender resulted in a camera that was placed correctly in relation to the remaining system parts.

Results from the post-processing scripts for simulating the fringing effect and the sensor noise are seen in *Figure 4.13* and *Figure 4.14* respectively. Notice the horizontal line structures in the ground truth sensor image of *Figure 4.14*.

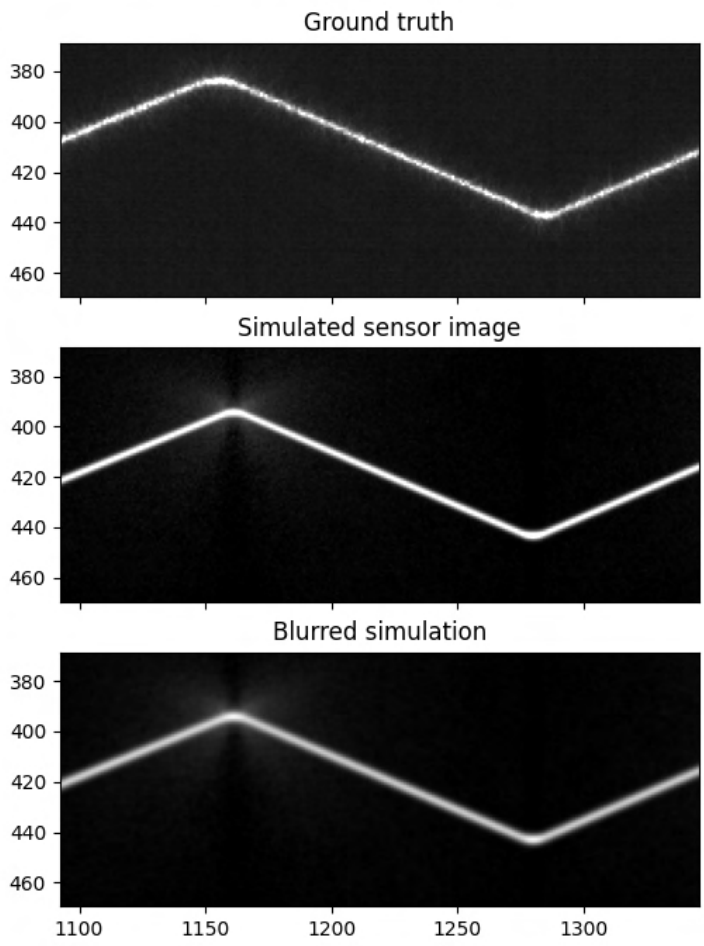


Figure 4.13: Result from the simulation of the unknown blurring/blooming/fringing effect, described in section 3.2.2, showing the simulated sensor image with (bottom) and without (middle) the applied effect, compared to the reference image (top). The laser line was simulated using the texture projector.

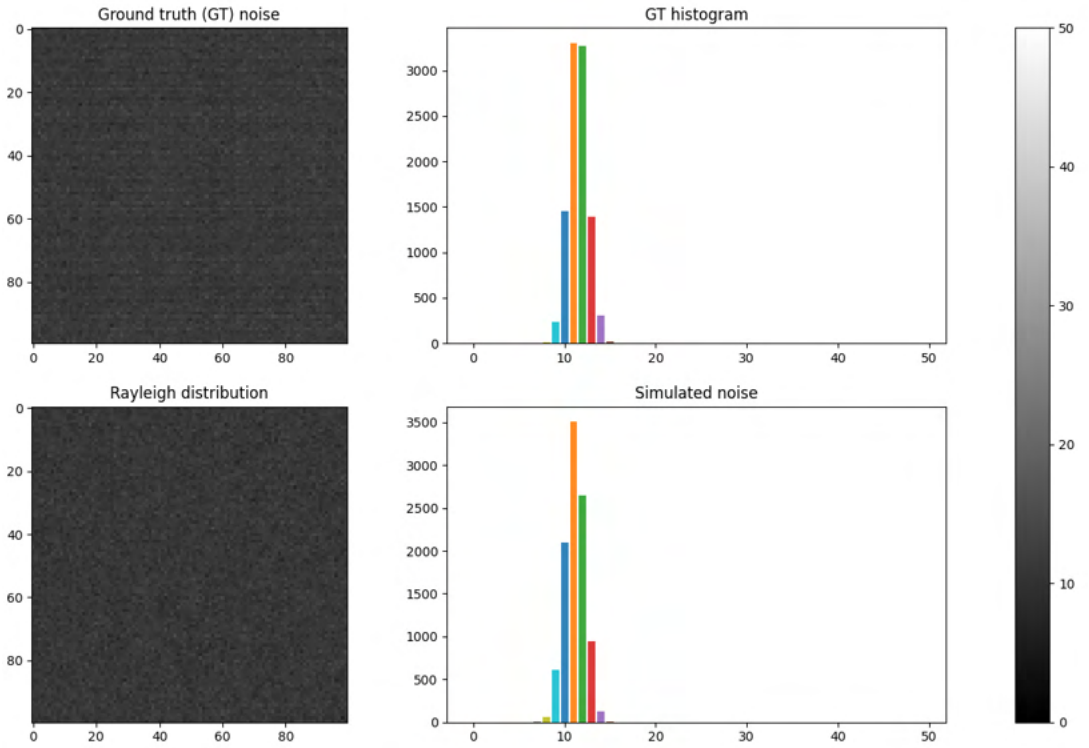


Figure 4.14: Results from the sensor noise simulation, presented in section 3.2.2, showing a close-up of a small and unlit region of the sensor images (left) together with their intensity distributions (right).

4.2.3. Measurement objects simulation

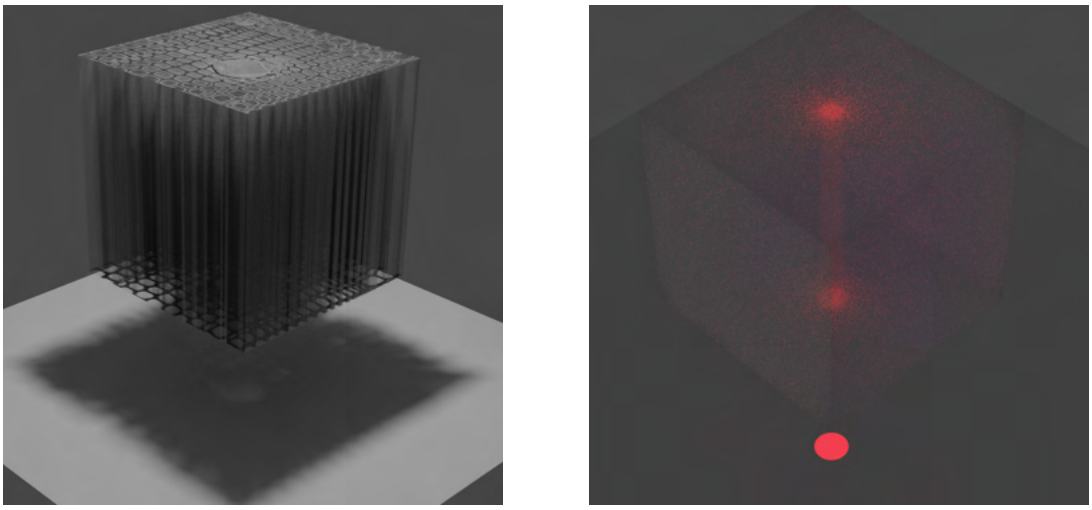
The following sections present the results of simulating a wooden plank and a blister package. Results from the various approaches for simulating anisotropic subsurface scattering are also presented.

Raw wooden plank

Figure 4.15 shows results from the wood simulation approach that was based on procedural textures. The method using microscopic images resulted in the renderings seen in *Figure 4.16*. Note how the red spot-lights in *Figures 4.15b* and *4.16b* maintain their circular shape on top of the wooden textures. Anisotropic subsurface scattering was not achieved using any of these methods.

(a) (b)

Figure 4.15: Results from procedural wood texture generation, in accordance with the method presented in section 3.2.3, showing a rendered image of a custom texture in (a), compared to a texture provided by BlenderKit [46] in (b). The spot-light in the centre of (b) is not affected by the texture as desired.



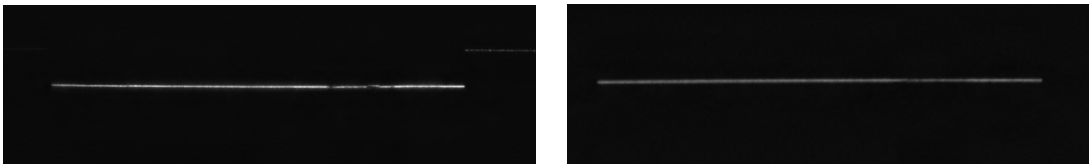
(a) (b)

Figure 4.16: Results from the wood texture approach that was based on microscopic images, showing the 2D-image texture transformed into 3D over a cube, before it was scaled to match the microscopic detail level (a). (b) depicts how the microscopic details does not affect the red spot-light to change shape; the light is uniformly scattered despite the hollow tracheids.

Figure 4.17: Rendered image of a scene with the projector laser line on top of the wooden plank that was simulated by UV-mapping a high resolution photograph onto the geometry.

Figure 4.17 presents the result from the approach of UV-mapping a wood texture onto a rectangle. To show the wooden texture, environmental lighting was included in the scene, withering the scattering effect of the texture projector laser somewhat.

Figure 4.18 shows the difference in the sensor image from the physical laser triangulation of a wooden plank and the result of the simulation of a sensor image. Both sensor images illustrate a laser line placed on top of the same knot of the wooden plank. The simulated sensor image was produced using the texture projector laser and the UV-mapping approach for the wood material.



(a) (b)

Figure 4.18: The reference sensor image from the physical laser triangulation session of a wooden plank (a), compared to the result of the simulation of sensor images in (b).

Blister package

The simulated blister package is depicted in *Figure 4.19*, showing the geometry and material lit by an environmental point-light, as well as by the texture projector laser.

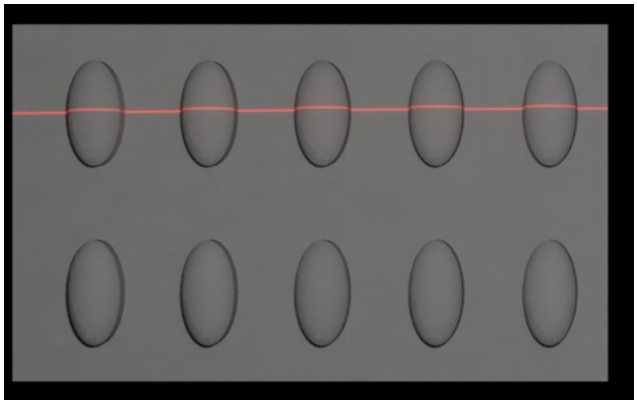
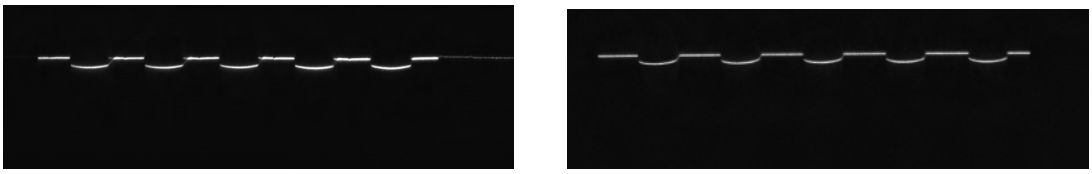


Figure 4.19: Rendered image of a scene with the projector laser line on top of the simulated blister package.

Figure 4.20 compares the sensor image from the physical laser triangulation with the result of the simulated sensor image of a blister package. The laser line is placed on the top row of blisters, where the leftmost blister does not contain a pill.

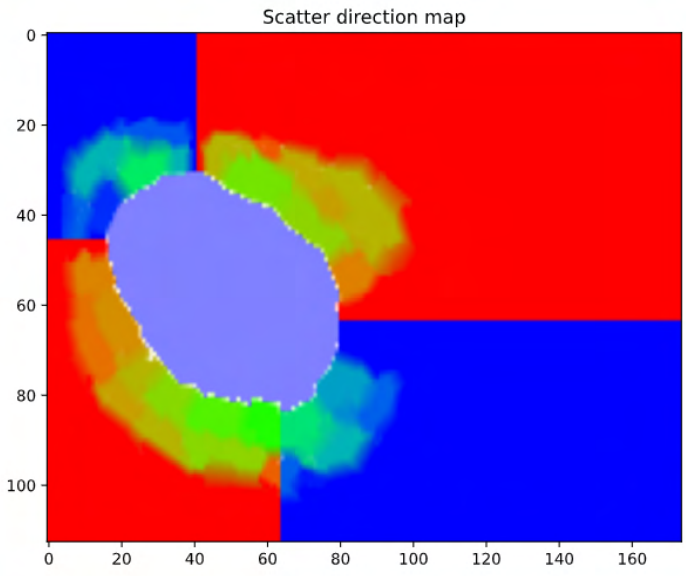


(a) (b)

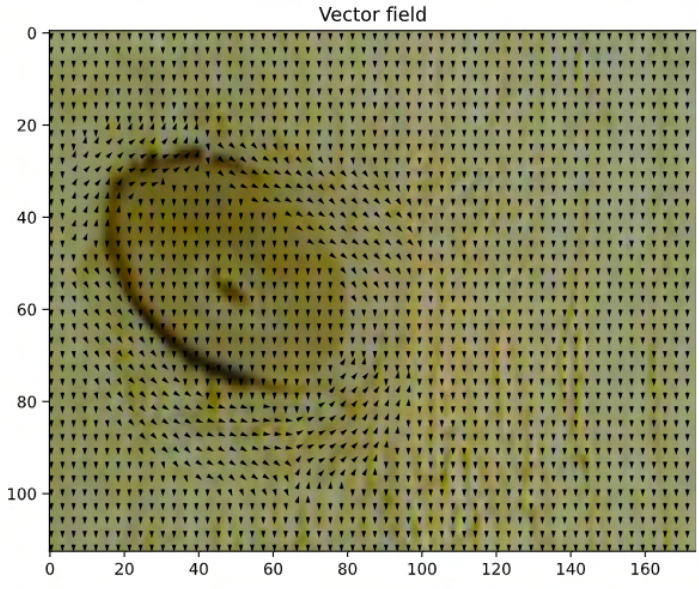
Figure 4.20: The reference sensor image from the physical laser triangulation session of a blister package (a), compared to the result of the simulation of sensor images in (b).

Anisotropic subsurface scattering

The subsurface scattering directions of the reference image of a wooden plank is illustrated in *Figure 4.21* below, showing both the initial colour direction map and the corresponding vector field. All regions of the colour map are fully opaque except for the knot-area that has a alpha-value of 0.5. The vector field is illustrated on top of a photograph of the reference plank.



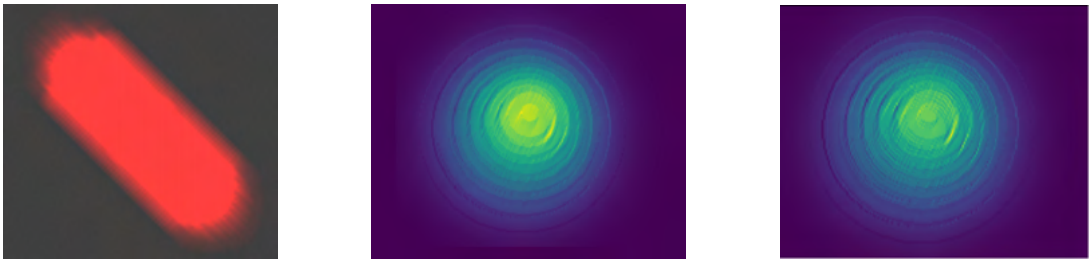
(a)



(b)

Figure 4.21: Result of the method for representing subsurface scattering directions of different regions of a wooden plank (background of (b)) as RGBA-colours (method presented in section 3.2.3). The coloured map illustrated in (a), together with the corresponding vector field that was obtained by translating the colour map to slopes in (b).

Figure 4.22 shows results of the anisotropic subsurface scattering script, presented in section 3.2.3, where the following line algorithms were used: linear equation, anti-aliasing lines and Bresenham's lines. *Figure 4.22a* depicts directed scattering along the ascending diagonal of a solid and red laser point. *b* and *c* illustrate scattering along the descending diagonal of a circular area with Gaussian distribution. The linear equation was used in *a*, anti-aliasing lines were used in *b* and the scattering of *c* was performed using Bresenham's lines. All three results include unwanted lines or streakiness due to the instability of the methods.



(a) (b) (c)

Figure 4.22: Results from the anisotropic subsurface scattering simulation using line algorithms, showing undesired streakiness due to instability. The algorithms used were the linear equation (a), anti-aliasing lines (b) and Bresenham's lines in (c), as presented in section 3.2.3.

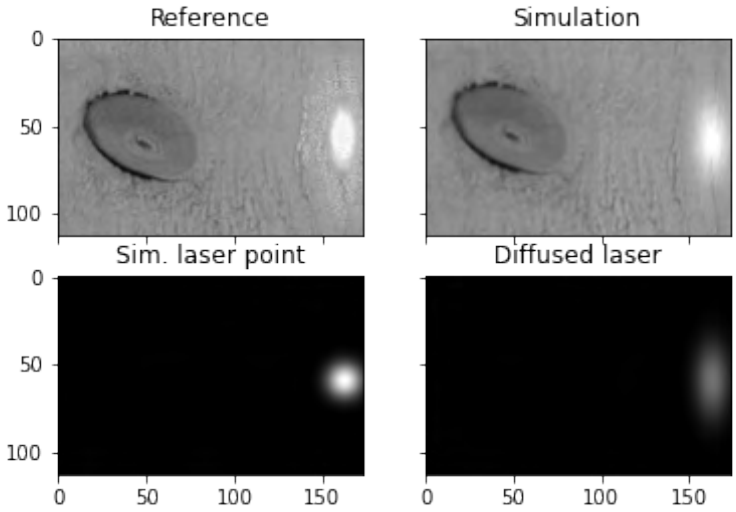


Figure 4.23: Result from the diffusion approximation process, presented in section 3.2.3, where a point laser was simulated as a filled circle with Gaussian intensity distribution and placed on an area without grain deviations (no knots). 20 iterations of diffusion were executed to spread the laser intensity according to the direction map shown in *Figure 3.10*.

The diffusion approach, presented in section 3.2.3, resulted in the images depicted in *Figures 4.23* and *4.24*, where the top rows compare the simulation results (right) with the reference images (left). The bottom rows of these figures illustrate the initial laser points before (left) and after (right) being diffused a given number of iterations. *Figure 4.23* shows the result of simulated anisotropic subsurface scattering on clear wood, whereas *Figure 4.24* illustrates the phenomenon when a point laser is lit on a knot of a wooden plank.

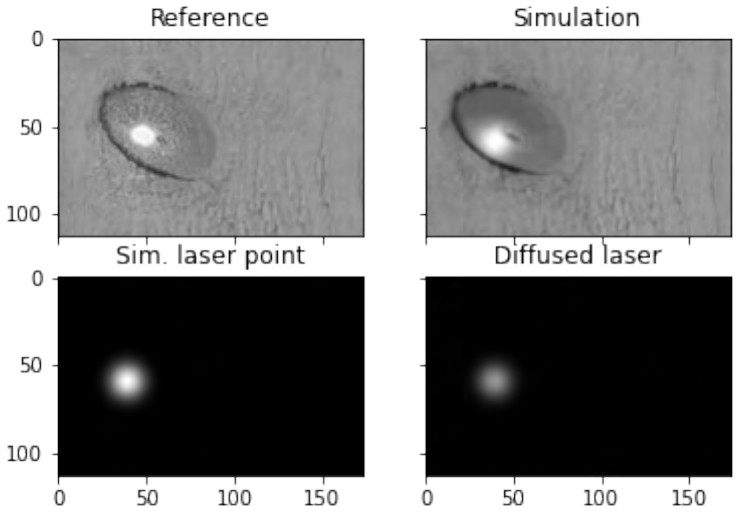


Figure 4.24: Result from the diffusion approximation process, section 3.2.3, where a point laser was simulated as a filled circle with Gaussian intensity distribution and placed on a knot area. 10 iterations of diffusion were executed to spread the laser intensity according to the direction map shown in *Figure 3.10*.

4.2.4. Post processing

The three post-processing scripts simulating speckle, blooming and sensor noise were applied in different orders, as presented in section 3.3. *Figure 4.25* compares the result of performing the blurring/blooming script first (followed by speckle and sensor noise), with the result of applying speckle first (followed by blurring and sensor noise).

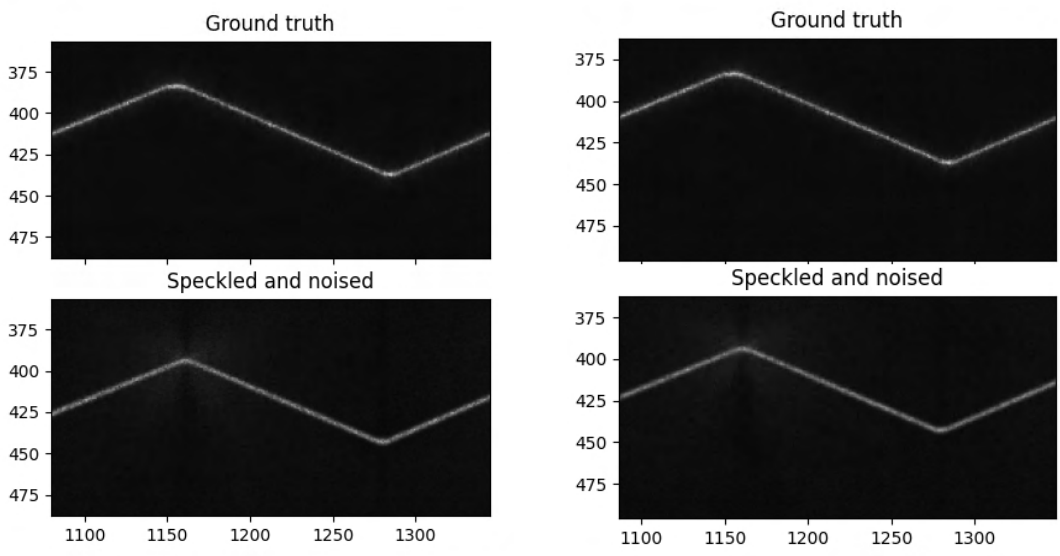


Figure 4.25: Results of applying the post-processing scripts in different orders, showing the order bloom - speckle - noise in (a), compared to the order speckle - bloom - noise in (b).

Scattering images

The scatter image from the physical laser triangulation is shown together with the simulated scatter image of a wooden plank in *Figure 4.26*. Knots and dirt are visible in the simulated scatter image. The rot along on the right side of the wooden plank is however barely visible in the simulated scatter image. *Figure 4.27* compares the simulated scattering image of a blister package with the scattering image from the practical laser triangulation. The figure shows a difference in the amount of scattered light in the empty and filled blisters, both in the ground truth and the simulated scatter images. The blisters where a pill is present are brighter than the hollow ones. As further seen in *Figure 4.27b*, the simulated blister shape differs from the ground truth and there are no specular reflections in the simulated image.

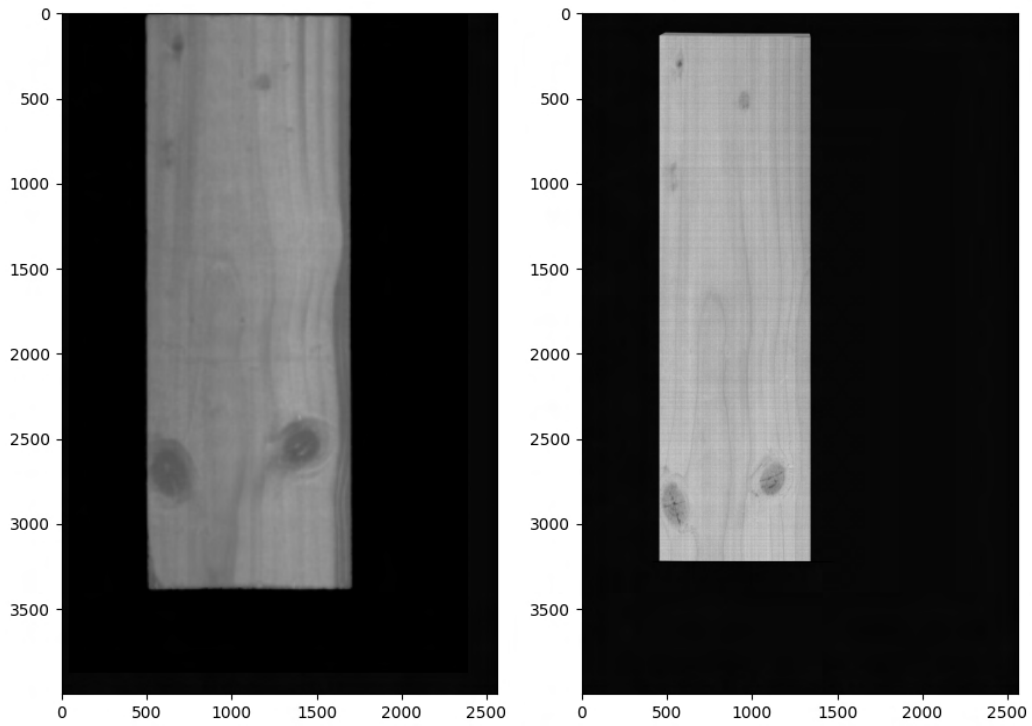
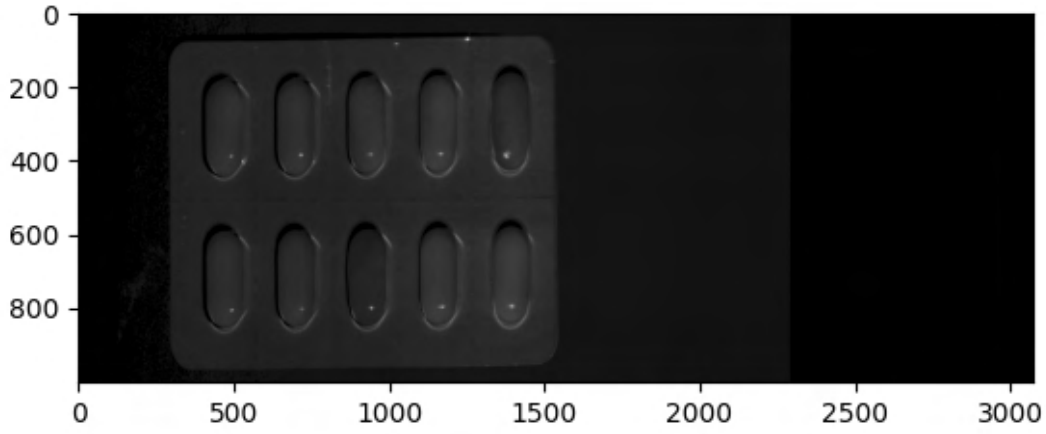
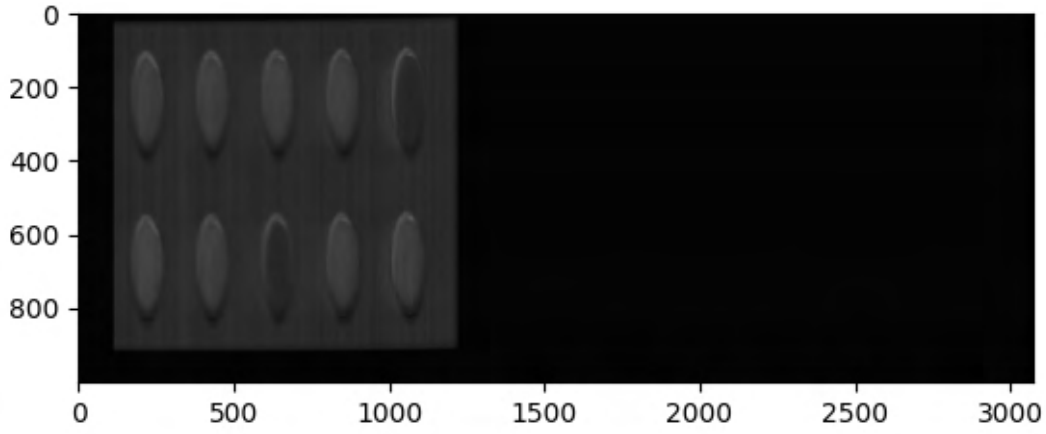


Figure 4.26: Reference scattering image from the physical laser triangulation session of a wooden plank (a), compared to the result of the scattering image simulation, presented in section 3.3.1, in (b).



(a)



(b)

Figure 4.27: Result from the simulation of scattering images, in accordance with the methodology presented in section 3.3.1, showing a reference image from the practical laser triangulation session of a blister package (a), compared to the simulated scattering image in (b).

5 Discussion

This chapter presents a discussion about the results covered in the previous chapter. Further, some of the used methods are discussed with hypotheses as to why they did not perform as expected, along with possible improvements.

5.1. Results

Most resulting images were presented highly magnified in this thesis. This was to show effects like scattering, sensor noise and the speckle patterns. Results that do not match the ground truth data at this level of detail may still be sufficiently realistic for many applications.

5.1.1. Laser line simulation

The results of the reshaped point-light, spot-light and area-light, depicted in *Figures 4.5* and *4.6* showed differences so small that they could be considered negligible. The same applies to the results of the two reshaping methods slit and shader nodes, seen in *Figure 4.4*; results with the same visual appearance could be obtained using both methods. For simplicity in the following discussions, these methods will all be referred to as reshaped light laser.

The texture projector laser was a better approximation of a Gaussian beam than the reshaped light laser. This was due to the blurred/faded edges, seen in *Figure 4.6d*. Because of this, the texture projector laser could perhaps be considered more realistic than the sharp and solid line of the reshaped light laser.

Due to it being a point laser, the volumetric Gaussian beam created with Cycles, shown in *Figure 4.7*, could not be considered finished for the aim of this thesis. It did possess the properties wanted for realistic laser simulation and could perhaps give satisfactory results if properly reshaped. The volumetric beam from LuxCoreRenderer, seen in *Figure 4.8* was not finalised either. Neither the varying thickness, nor the Gaussian intensity distribution was achieved. A cylinder lens was however proved to be a working method for reshaping point lasers, as seen in *Figure 4.9*.

When applying speckle onto a reshaped light laser and a projector laser there was a small difference in the brightness of the laser lines as seen in *Figure 4.12*. This was expected since the Gaussian blurring of the projector laser distributed the intensity over a larger area, making

the line appear darker. Using Rayleigh intensity distribution resulted in a brighter speckle pattern. This was expected since the sum of the Rayleigh histogram, presented in *Figure 4.10*, was larger than the sum of the exponential intensity distribution histogram. *Figure 4.11* shows that the appearance of the laser lines differed slightly depending on which intensity distribution the applied speckle patterns had.

5.1.2. Camera simulation

Figures 4.11 and *4.12* demonstrate effects that were likely caused by using focus at infinity and neglecting the Scheimpflug-adapter. A Scheimpflug-adapter together with a lens focused at infinity is expected to decrease the field of view, which in turn should magnify the depicted object. As seen in these figures, the perspective seemed to differ between the simulated and the ground truth images. The ground truth images appear slightly more zoomed-in or magnified. Simulation of the fringing/blooming effect as Gaussian blur did not result in the hairy/spiky phenomenon that was seen in the ground truth images, see *Figure 4.13*. This was expected and considered as a sufficiently good approximation for this thesis.

In the comparison of close-ups of ground truth and simulated sensor images, seen in *Figure 4.14*, horizontal lines or structures were seen in an unlit area of the ground truth data. These effects were not considered when simulating the sensor noise and thus not found in the resulting image. The histograms show that the simulated sensor images had a level of noise that matched the reference data, despite the absence of horizontal lines.

5.1.3. Wooden plank simulation

The simulated wood material that best resembled the reference plank visually was the one created through UV-mapping, seen in *Figure 4.17*. This was because the desired surface properties, like knots, rays and rot, were included and placed in accordance with the reference plank. The UV-mapped wood material had different scattering properties depending on whether the laser line was on top of the clear wood or a knot. The cause of the differed scattering is likely because of the darker colour of the knot.

The simulated sensor images of the wood were darker than the ground truth sensor images, as seen in *Figure 4.18*. Despite this, the simulated scattering image was brighter than the one obtained from physical laser triangulation, as seen in 4.26. This was not expected and the cause for this is unknown.

5.1.4. Blister package simulation

The modelled geometry of the blister package did not match the reference object properly, as seen when comparing *Figures 3.3* and *4.19*. The simulated blisters were slightly too rounded and they did not contain the small bumps and crevices seen in the reference package. The simplifications of the simulated geometry were seen in the differences between the simulated and the ground truth sensor images, presented in *Figure 4.20*. Because of this simplification the laser line of the simulated image may be considered too intact or ideal to be a proper representation of the ground truth data. The intensity of the laser line was also lower in the simulation result than in the results from the physical laser triangulation. The scattering difference between empty and filled blisters was minimal and hard to see in both the ground truth and simulated sensor images.

The blister packages, both ground truth and simulated, were elongated in the direction of displacement in the scattering images, seen in *Figure 4.27*. This was not expected, not wanted and possibly made the blister shape differences even clearer. The scattering images also illustrated that the simulated package material had less specular reflectance than the reference package. The difference between empty and filled blisters in the simulated scattering image was however similar to the ground truth image.

5.1.5. Post-processing

In the comparison of the post-processing script orders, seen in *Figure 4.25*, one resulting laser line appeared darker and/or smoother than the other. When performing the blurring/fringing script after the speckle script, the applied speckle pattern was blurred. This caused the brighter speckles to be softened, thus resulting in a laser line with flattened intensity. The intensity of the second result, where speckle was applied after blurring, was closer to ground truth laser line. This script order (blur - speckle - noise) could therefore be considered the best.

5.2. Method

The following sections include discussions of the chosen methodology for simulating the system parts of a laser triangulation set-up. A validation analysis along with possible improvements is also presented.

5.2.1. Practical laser triangulation

Since the practical laser triangulation result images were (partly) overexposed, they were not suitable for use as ground truth data. Not ideal and unrealistic reference images likely result in simulations that are, at a minimum, just as flawed as the ground truth. The data of the saw-tooth object was better suited for usage as ground truth, since it was properly exposed and contained the interesting phenomena.

5.2.2. Laser line simulation

Creating a slit or opening between two geometries was a time efficient and easy to set up method. It did not require any mathematical computations but it could clutter the scene somewhat, if it already contained many elements. Using shader nodes to reshape a light into a beam took more time to implement than the slit-method, due to the building of the node tree. Expressing the required maths with a node tree was perhaps not as straight forward as creating and placing two geometries.

The results from the finalised laser line methods proved that the texture projector was the best choice in the following aspects: it was fast and easy to set up and the image texture could be altered to include the wanted laser properties. Nevertheless, all laser methods could potentially be used to produce similar results, by varying the parameters of the post-processing scripts. The blurred edges of the projector laser could possibly be obtained for the reshaped light lasers as well. For example, a bigger blurring radius could be used in the blooming script, resulting in stronger fading of the edges. If there would still be a difference in intensity between the two laser methods after this, a smaller speckle scaling factor could be used for the reshaped light laser to darken the result. However, the post-processing scripts affect the measurement object as well as the laser line. Using a larger blurring radius would cause sharp edges of the object to lose their sharpness, possibly lowering the realism of the simulated object. This further adds to the advantages of using the texture projector laser; the post-processing required for this method has less effect on the simulated measurement object.

The desired laser property of varying thickness could potentially be added to the reshaped light lasers. For the slit method, the geometries could be shaped and placed so that the thickness of the slit varied. Using shader nodes, the thickness of the beam could be defined to vary depending on the texture coordinates. For the shader node approaches, the same texture coordinates could also be used to vary the intensity of the light beam. Achieving intensity variations with the slit method would perhaps not be as straight forward. The transparency of the slit-creating geometries could potentially be varied to successively let through more or

less light. All of these approaches would however most likely be more complicated to implement than the texture projector laser. A new image texture with varying thickness could be generated for the projector, by gradually adding thicker lines towards the ends of the laser, before applying the Gaussian blur. After studying ground truth lasers in large magnification, the variations in beam thickness could however be considered to be insignificantly small.

The extent of this thesis was not enough for the research needed in order to properly familiarise with LuxCoreRenderer. The Gaussian beam that was approximated using this renderer could potentially be developed to a highly realistic and physically accurate laser line. Comparisons of such a line with the texture projector laser would be interesting to perform.

Speckle theory stated that the intensity distribution of a speckle pattern should be exponential, as presented in sections 2.3.2 and 3.2.1. Despite this, the results of *Figure 4.11* suggest that Rayleigh-distributed speckle patterns also could be used to simulate laser lines that resemble ground truth data. Patterns of either of the two distributions require intensity scaling to match the reference laser. The fact that a scaling factor was needed can be seen to reduce the physical accuracy of the method.

5.2.3. Camera simulation

Since the simulation was performed in Blender and because the software included relatively few camera properties, the realism of the camera simulation was also limited. By using infinite focus distance, the entire Blender scene was rendered sharply in a not physically accurate manner. If another rendering software was used, that allowed for more parameter specifications, the simulation could potentially have been more alike the physical camera. The perspective, FoV and magnification of the ground truth image could perhaps have been obtained if a Scheimpflug-adapter was simulated and if the focus was set to match the settings of the practical laser triangulation set-up.

The bloom/fringe effect that occurred on the physical laser line could be simulated to better resemble ground truth images. The effect was probably caused by different aberrations in the camera and/or sensor. The fringing could be approximated in a number of alternative ways rather than as a Gaussian blur. One possible method would be to use some kernel, to achieve different blurring directions depending on the laser line angle. This would result in a laser line more alike the one used in the physical laser triangulation. There is further a node in Blender to simulate camera distortions and aberrations. It is called the *Lens Distortion Node* and it could be included in the compositing step to add effects like jitter and chromatic aberrations, possibly resulting in a more realistic fringing effect [51]. Since this thesis mainly focused on scatter and speckle, realistic simulation of this blooming/fringing effect was considered out of the scope. The sensor noise simulation could perhaps be improved by performing a frequency analysis. This was however considered to be too detailed for the extent of the thesis.

5.2.4. Measurement objects simulation

The procedural methods used to simulate wood texture was proven efficient but impractical since the shape of the laser was not altered, as seen in *Figure 4.15*. The procedural texture was also hard to modify to replicate the wooden plank used in the physical laser triangulation. All properties of the physical plank were not generated with the procedural method. Rot, for example, was excluded since it made the node tree in Blender too complex.

The final approach, which was to UV-map a photograph of a wooden plank, resulted in scatter and sensor images that were similar to the ground truth data. As mentioned in the method section, the geometry of the wooden plank was modelled as a perfect 3D-rectangle. The reference plank however, did not have perfectly straight sides and edges. When UV-mapping a photograph of this not perfectly rectangular plank onto the ideal geometry, parts

of the photograph was removed. This caused some differences in the ground truth and simulated results.

Custom shaders can be implemented in Blender using the Open Shading Language (OSL) [52]. A custom BSSRDF could potentially be produced to simulate a realistic wooden material. However, OSL currently only supports rendering on the CPU, making such an approach unsuitable; rendering thousands of profiles on the CPU would take several days.

As mentioned earlier, the blister geometry and material had some flaws. If the geometry was fine-tuned with more details and the material included realistic properties such as specular reflectance, the sensor images would perhaps look more similar to the ground truth. The main focus of the blister package simulation was, as mentioned, to obtain a scattering difference between hollow and filled blisters that matched reference data. Because of this, producing a highly detailed geometry together with a more realistic material was not a top priority.

5.2.5. Anisotropic subsurface scattering simulation

The creation of the vector field representation for scattering directions was time consuming. To manually create gradients and interpolating between different colours/direction was a tedious work process. If the vector field instead was described by mathematical equations or if the interpolations were done according to a script, the process could perhaps be sped up and simplified.

The undesired streakiness that appeared in the line algorithms could potentially be minimised to sufficiently low levels by blurring the resulting images. The line algorithms will however lack energy conservation properties compared to the diffusion approximation.

A possible solution for simulating anisotropic subsurface scattering in Blender would be to extend or rework the open source implementation of the subsurface scattering shader node. There is also the option to program a custom renderer for Blender that include path tracing for anisotropic subsurface scattering. However, that alone would probably be a thesis work. Alternatively another rendering software could be tested to simulate the phenomena.

5.2.6. Post-processing

Each post-processing script depended on a number of parameters that could be tweaked to possibly improve the realism of the results. The sensor noise parameter values were however based on measurements in ground truth sensor images and could therefore be considered appropriate. The speckle parameters pattern size, circular mask diameter and scale factor could be assumed to have a large impact on the resulting laser lines, due to their connection to speckle size and the overall intensity of the patterns. More thorough testing should be performed to optimise these parameter values.

The numbers of profiles for each scattering image, along with the displacement between each profile were obtained from and matched to the parameters of the physical laser triangulation session, as mentioned in sections 3.1 and 3.3. Identical laser centre extraction algorithms were used for the simulated and ground truth sensor images. The same applies for the scattering measurement algorithm; the same methodology and parameter values were used for both sessions. The realism of the simulated scattering images should therefore only depend on the quality of the simulated sensor images.

5.2.7. Comparison and evaluation

All evaluations of this thesis work were purely subjective, since they were performed by the thesis workers alone. Further, the screen monitor used for comparing the simulation results with ground truth data affected the evaluations. This was because various screen resolutions were able to present different levels of detail; some effects were not possible to see on all the

screens used during the thesis work. Since not all result images were viewed on different screens, anomalies and unwanted effects could have been missed.

Thorough user test should be performed to validate the results and discussions presented. Due to the relatively detailed level of theory researched in this thesis, qualitative testing would perhaps be more efficient than quantitative. Background knowledge is most likely needed for a tester to be able to compare the sensor and scattering images properly.

5.2.8. Source criticism

The majority of articles, papers and reports used in this thesis work were peer-reviewed. For the references that deviated from this, other peer-reviewed sources, on the same topic, were used to support the theory and/or method in question.

6 Conclusion

The aim of this thesis was to examine methods for realistically simulating each system part of a laser triangulation set-up. Simulated measurement objects were supposed to include the subsurface scattering phenomenon and the simulation results should be validated through comparison with ground truth data.

Various methods for simulating system parts have been presented together with results and possible improvements. These include three laser line simulation approaches, a camera simulation, three methods for wood simulation and one approach for simulating blister packages. Materials with varying subsurface scattering were constructed for the blister package simulation. To extensively explore subsurface scattering of wood, an isolated algorithm was implemented to anisotropically diffuse a simulated point laser. All resulting images were compared to corresponding ground truth data.

The texture projector method was the best out of the three tested laser line approaches, because it was simple to set up and since it gave realistic results. Blender Cycles was proven to not be able to reshape a simulated volumetric Gaussian laser beam using a cylinder lens. LuxCoreRenderer allows reshaping of a volumetric beam with the usage of a cylinder lens, but the Cycles shader node tree was not successfully translated into LuxCoreRenderer texture nodes. Results of the camera simulation suffered from not including a Scheimpflug-adapter as well as from assuming infinite focus. For the wooden material simulation, the results of the UV-mapping approach best resembled the reference plank. This method was also the most straight forward to implement. None of the simulated wooden materials caused light to scatter anisotropically. Because of this, the script for simulating anisotropic subsurface scattering was implemented. The representation of scattering directions, through colour mapping, was inefficient and not very accurate, due to it being roughly drawn by hand. The blister package simulation resulted in a scattering image with clear visible difference between the hollow and filled blisters.

6.1. Research questions

The research questions of the thesis could be answered as follows:

1. How can realistic line lasers be simulated in the rendering software Blender? Line lasers can be simulated in Blender using various approaches, as presented throughout this report. The built-in light sources can be reshaped with shader node trees or by creating thin openings between planes. These approaches result in laser lines with constant thickness and intensity. Blur and speckle patterns can be added through post-processing scripts, which may cause these laser lines to achieve relatively high realism.

Line laser textures can be included via the *Image texture* node. These textures can then be projected through a built-in light source to generate laser lines. The realism of such a resulting laser line depends mostly on the realism of the laser texture. With a texture containing the main characteristics of a real laser, the texture projector laser can be used to simulate laser lines of high realism.

A point laser can be generated as a geometry with an emitting volume material. The material can be implemented using Cycles shader nodes to approximate the irradiance of a Gaussian beam. Point lasers can be reshaped into line lasers using a cylinder lens. Creating realistic cylinder lenses is currently not possible using Cycles renderer. However, LuxCoreRenderer includes better caustics, enabling for creation of cylinder lenses. The Gaussian beam approximation can potentially be implemented using this renderer as well. Combined with the working cylinder lens, this would result in a realistic line laser.

2. How can realistic wood and blister package materials that include subsurface scattering be simulated in Blender? Anisotropic subsurface scattering can currently not be simulated in Blender. Uniform scattering can be applied to materials through various shader nodes, such as the *Subsurface scattering* and *Volume scatter* nodes. Blister package materials can be simulated, for example, in accordance with the node trees presented in *Figure 3.12*. Parameters such as specular reflectance should however be tweaked in order to obtain higher realism. Realistically appearing wooden materials are efficiently created through UV-mapping high resolution photographs. Wooden materials that affect light truthfully are not possible to simulate in Blender due to the limited subsurface scattering support/implementation.

3. How can the realism of simulated laser triangulation be evaluated through comparison with practical laser triangulation? Evaluation of each system part isolated is not possible. Dimensions, colours, power and other properties are simulated to match the settings of the physical system parts. To evaluate the realism of this parameter matching, the only data available are the captured or simulated result images. Sensor images are created during a physical laser triangulation session. In post-processing, these images or profiles can be combined to generate scattering images. The simulated laser triangulation set-up can be evaluated by comparing the simulated and ground truth sensor images. Looking at close-ups of these images, factors and occurrence of phenomena like intensity, scattering, speckle and blur can be used to determine the quality of the simulation. The smaller visual differences between the sensor images, the better is the simulation. The simulated sensor images can also be combined to scattering images. Comparison of the simulated and ground truth scattering images can also be used to evaluate the simulated laser triangulation system.

6.2. Future work

Future work of this thesis includes finalising the approximation of a Gaussian beam using LuxCoreRenderer. With more knowledge about the LuxCoreRenderer, the shader node tree that was built in Cycles should be properly translated into a volumetric texture supported by the LuxCoreRenderer. The image texture for the projector laser should also be improved to include more laser properties. Local variations in wavelength and focus, as well as varying

thickness through a beam waist, should be introduced to the image texture to enhance the realism of the resulting laser.

The blooming/fringing effect should be properly investigated to allow for realistic simulation. Further, exploration of alternative methods for efficient vector field generation is highly interesting. A more automated algorithm would be desired that could translated an image of a measurement object into a vector field representation of the object's scattering directions. Combinations of various image processing approaches like edge detection and Hough transform could be tested and evaluated.

Other approaches for simulating subsurface scattering of a point laser in wood should also be researched, perhaps using other rendering software or by implementing a custom path-tracer. The diffusion algorithm for simulating anisotropic subsurface scattering should be extended from the 2D image/sensor space to the three dimensional object space. Subsurface scattering is naturally a 3D phenomenon that cannot be realistically simulated by a 2D diffusion process. Extensive user tests should further be performed to acquire objective results.

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Appendix A

Speckle pattern coordinate

According to Bergmann et al., the third pattern coordinate u was derived as follows:

$$u = \left(\frac{|a_p^c + a_p^s|}{D} + \frac{dz}{dz_0} \right) \times M, \tag{A.1}$$

where the parameters were defined as:

- a_p^c: camera movement
- a_p^s: aperture displacement caused by surface/object movement
- $\frac{d_z}{d_{z0}}$: change in distance between camera and surface
- D: lens diameter
- M: half the number of patterns.

The practical and simulated laser triangulation system only had movement in one dimension and the measurement object was the only system part that moved. Therefore, $a_p^c = \frac{d_z}{d_{z0}} = 0$, allowing for equation A.1 to be rewritten as follows:

$$u = \frac{|a_p^s|}{D} \times M. \tag{A.2}$$

a_p^s is divided into two components $a_{p,x}^s$ and $a_{p,y}^s$, defined as follows:

$$\begin{aligned} a_{p,x}^s &= -\frac{d_{SP}}{cos(q_1)} \left(a_x \left(\frac{l_{sx}^2 - 1}{d_{SL}} + \frac{l_x^2 - 1}{d_{SP}} \right) + a_y \left(\frac{l_{sx} l_{sy}}{d_{SL}} + \frac{l_x l_y}{d_{SP}} \right) + a_z \left(\frac{l_{sx} l_{sz}}{d_{SL}} + \frac{l_x l_z}{d_{SP}} \right) \right) \\ &+ d_{SP} \frac{sin(q_1) sin(q_2)}{cos(q1) cos(q_2)} \left(a_x \left(\frac{l_{sx} l_{sy}}{d_{SL}} + \frac{l_x l_y}{d_{SP}} \right) + a_y \left(\frac{l_{sy}^2 - 1}{d_{SL}} + \frac{l_y^2 - 1}{d_{SP}} \right) + a_z \left(\frac{l_{sy} l_{sz}}{d_{SL}} + \frac{l_y l_z}{d_{SP}} \right) \right) \\ a_{p,y}^s &= -\frac{d_{SP}}{cos(q_2)} \left(a_x \left(\frac{l_{sx} l_{sy}}{d_{SL}} + \frac{l_x l_y}{d_{SP}} \right) + a_y \left(\frac{l_{sy}^2 - 1}{d_{SL}} + \frac{l_y^2 - 1}{d_{SP}} \right) + a_z \left(\frac{l_{sy} l_{sz}}{d_{SL}} + \frac{l_y l_z}{d_{SP}} \right) \right). \end{aligned} \tag{A.3}$$

The parameters of these equations were defined as:

- q_1: rotation around the x-axis of the surface coordinate system (to get the first axis of the sensor coordinate system)
- q_2: rotation around the first axis of the intermediate coordinate system
- $a(a_x, a_y, a_z)$: surface displacement
- $l(l_x, l_y, l_z)$: sensor normal (in surface coordinate system)

- $l_s(l_{sx}, l_{sy}, l_{sz})$: normalised vector between surface and laser (in surface coordinate system)
- d_{SP}: distance between surface and pupil/lens
- d_{SL}: distance between surface and laser.

Since the object only moved along the y-axis, $a_x = a_z = 0$. l_s was known to be $(0, 0, 1)$ and $q_2 = 0$. Equation A.3 could therefore be simplified as follows:

$$\begin{aligned} a^s_{p,x} &= -\frac{a_y l_x l_y}{cos(q_1)} \\ a^s_{p,y} &= -d_{SP} \times a_y \left(\frac{-1}{d_{SL}} + \frac{l_y^2 - 1}{d_{SP}} \right). \end{aligned} \tag{A.4}$$

Appendix B

Division of work - responsibilities

Table B.1: Division of responsibilities

Task	Head responsible
Laser simulation methods	
Spot-light:	HK[1]
Point-light:	HK
Area-light:	SK[2]
Texture projector:	SK
Gaussian beam in Cycles:	SK
Gaussian beam in LuxCoreRenderer:	HK
Simulation of measurement objects	
Wood - volumetric material in Cycles	SK
Wood - volumetric material in LuxCoreRenderer	HK
Wood - procedural textures	SK
Wood - UV-mapping photograph	HK
Blister - geometry	SK
Blister - material	HK
Profile rendering for scatter images	
Wood	SK
Blister	HK
Rendering and post-processing scripts	
Scenes and scripts for rendering	SK
Blurring simulation	HK
Speckle pattern simulation	HK
Sensor noise simulation	SK
Simulated anisotropic subsurface scattering	HK and SK
Comparisons	
Laser methods and post-processing scripts	HK
Scatter images	SK

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